

Representation of Distributed Polynomials

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1 The polyopvar Representation

For $\mathbf{x} = \begin{bmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_p \end{bmatrix}$ with $\mathbf{x}_i \in L_2[\Omega_i]$, we consider a class of scalar-valued distributed polynomials in $\mathbf{x}$, taking the form

$$\mathbf{p}(\mathbf{x}) = \mathcal{C}\mathbf{Z}(\mathbf{x})$$

where $\mathbf{Z}(\mathbf{x})$ is a basis of distributed monomials in $\mathbf{x}$, and where $\mathcal{C}$ is a row vector of operators acting on the basis. In particular, we let

$$\mathcal{C} = [\mathcal{C}_1 \quad \mathcal{C}_2 \quad \cdots \quad \mathcal{C}_{n_Z}], \quad \mathbf{Z}(\mathbf{x}) = \begin{bmatrix} \mathbf{x}_1^{\otimes d_{1,1}} \otimes \cdots \otimes \mathbf{x}_p^{\otimes d_{1,p}} \\ \mathbf{x}_1^{\otimes d_{2,1}} \otimes \cdots \otimes \mathbf{x}_p^{\otimes d_{2,p}} \\ \vdots \\ \mathbf{x}_1^{\otimes d_{n_Z,1}} \otimes \cdots \otimes \mathbf{x}_p^{\otimes d_{n_Z,p}} \end{bmatrix},$$

where $n_Z \in \mathbb{N}$ is the number of monomials in the basis, and where $\otimes$ denotes the tensor product. We represent this class of distributed polynomials as **polyopvar** objects, taking the following form

- **C** a **tensorvar** object representing the operator $\mathcal{C}$ acting on the distributed monomial basis $\mathbf{Z}(\mathbf{x})$.
- **degmat** $n_Z \times p$ array of integers, specifying the degrees of the monomials appearing in the basis $\mathbf{Z}(\mathbf{x})$, so that $d_{i,j} = \text{degmat}(\mathbf{i}, \mathbf{j})$.
- **varname** $p \times 1$ cellstr specifying the names of the state variables $\mathbf{x} = \begin{bmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_p \end{bmatrix}$ in which the distributed polynomial is defined.
- **pvarname** $N \times 1$ cellstr specifying the full list of names of the spatial variables, $s = (s_1, \dots, s_N)$, on which the state variables are defined.
- **dom** $N \times 2$ array specifying for each of the spatial variables s_j for $j \in \{1, \dots, N\}$ the domain $[a_j, b_j] = \text{dom}(\mathbf{j}, :)$ on which this spatial variable is defined.
- **varmat** $p \times N$ logical array, with the element **varmat**(**i**, **j**) for $i \in \{1, \dots, p\}$ and $j \in \{1, \dots, N\}$ specifying whether (1) or not (0) state variable $\mathbf{x}_i$ depends on spatial variable s_j .

1.1 The tensorvar Structure

In the **polyopvar** structure, the coefficient operator $\mathcal{C}$ acting on the distributed monomial basis is represented by a **tensorvar** object **C**. This structure has just a single field

- **ops** $1 \times n$ cell, with the i th element representing the coefficient operator $\mathcal{C}_i$ acting on the i th distributed monomial.

In this structure, the different elements **C.ops**{**i**} may represent different types of operators, represented using different structures. In particular, we distinguish between tensor-PI operators and functional operators.

1.1.1 Tensor-PI operators

The class of tensor-PI operators is used to represent elementwise products of state variables, mapping elements of $L_2[\Omega^d]$ to elements of $L_2[\Omega]$. This class of operators is used to represent distributed polynomial vector fields in the PIE representation. We define a general class of tensor-PI operators as follows.

Definition 1. For given domains $\Omega, \Omega_1, \dots, \Omega_d$ we say that $\mathcal{P} : L_2[\Omega_1 \times \dots \times \Omega_d] \rightarrow L_2[\Omega]$ is a tensor-PI operator if there exists $m \in \mathbb{N}$ and PI operators $\mathcal{R}_i : L_2[\Omega_i] \rightarrow L_2[\Omega]$ such that

$$(\mathcal{P}(\mathbf{x}_1 \otimes \dots \otimes \mathbf{x}_d))(s) = \sum_{j=1}^m (\mathcal{R}_{j,1}\mathbf{x}_1)(s) \cdots (\mathcal{R}_{j,d}\mathbf{x}_d)(s), \quad \forall \mathbf{x}_i \in L_2[\Omega_i], s \in \Omega. \quad (1)$$

This definition essentially includes all operators that can be represented as linear combinations of “tensor products” of PI operators¹, $\mathcal{P} = \sum_{i=1}^m \mathcal{R}_{i,1} \otimes \dots \otimes \mathcal{R}_{i,d}$. Here, the operators $\mathcal{R}_{i,j}$ are allowed to map from different function spaces, so long as they all map to $L_2[\Omega]$. For the purpose of representing a 1D PDE, this simplifies to $\Omega = \Omega_i = [a, b]$, and we get the following definition

Definition 2. For given $[a, b] \in \mathbb{R}$ and $d \in \mathbb{N}$, we say that $\mathcal{P} : L_2[[a, b]^d] \rightarrow L_2[a, b]$ is a tensor-PI operator if there exists $m \in \mathbb{N}$ and 3-PI operators $\mathcal{R}_i : L_2[a, b] \rightarrow L_2[a, b]$ such that

$$(\mathcal{P}(\mathbf{x}_1 \otimes \dots \otimes \mathbf{x}_d))(s) = \sum_{j=1}^m (\mathcal{R}_{j,1}\mathbf{x}_1)(s) \cdots (\mathcal{R}_{j,d}\mathbf{x}_d)(s), \quad \forall \mathbf{x}_i \in L_2[\Omega_i], s \in \Omega. \quad (2)$$

In the tensor-PI structure, a tensor-PI operator $\mathcal{P} := \sum_{i=1}^m \mathcal{R}_{i,1} \otimes \dots \otimes \mathcal{R}_{i,d}$ is represented as a $m \times d$ cell, with the element in row i and column j a `nopvar` object representing the operator $\mathcal{R}_{i,j}$. It is important here that all operators map to the same function space.

Note that, for the tensor product of 2-PI operators, we can parameterize a more general class of “tensor-PI” operators by 2^d polynomials $\mathbf{P}_\alpha \in \mathbb{R}[s, \theta]$

$$(\mathcal{P}\mathbf{w})(s) = \sum_{\alpha \in \{-1, 1\}^d} \mathbf{I}_\alpha(s, \theta) \mathbf{P}_\alpha(\theta) \mathbf{w}(\theta) d\theta, \quad \forall \mathbf{w} \in L_2[[a, b]^d], s \in [a, b], \quad (3)$$

where $\mathbf{I}_\alpha(s, \theta) = \prod_{i=1}^d \mathbf{I}(\alpha_i[s - \theta_i])$, for $\mathbf{I}(x - y) := \begin{cases} 1, & y \leq x, \\ 0, & y > x, \end{cases}$ the standard indicator function. This representation includes any linear combination of tensor products of PI operators, removing the need for the explicit sum in the definition. However, the current `tensopvar` structure does not support this representation.

1.1.2 Functional Operators

Functional operators map function-valued objects to real scalar values. These functionals are used to represent distributed polynomial functionals, such as the Lyapunov functional and its temporal derivative. Although such functionals $\mathcal{K} : L_2[\Omega] \rightarrow \mathbb{R}$ can always be expressed as an integral, $\mathcal{K}\mathbf{w} := \int_\Omega \mathbf{K}(\theta) \mathbf{w}(\theta) d\theta$, the kernel of this integral need not be polynomial. For our purposes, we will allow the kernel to be polynomial semi-separable, allowing different polynomial kernels depending on the ordering of the variables $\theta_1, \dots, \theta_d$. In the 1D case, we define the class of PI functionals as follows

Definition 3. For $[a, b] \subseteq \mathbb{R}$ and $d \in \mathbb{N}$, we say that $\mathcal{K} : L_2[[a, b]^d] \rightarrow \mathbb{R}$ is a PI functional if there exist $\mathbf{K}_\alpha \in \mathbb{R}[\theta_1, \dots, \theta_d]$ for $\alpha \in A_d$ such that

$$\begin{aligned} \mathcal{K}\mathbf{w} &= \sum_{\alpha \in A_d} \int_a^b \cdots \int_a^b \mathbb{I}_\alpha(\theta_1, \dots, \theta_d) \mathbf{K}_\alpha(\theta_1, \dots, \theta_d) \mathbf{w}(\theta_1, \dots, \theta_d) d\theta_d \cdots d\theta_1 \\ &= \sum_{\alpha \in A_d} \int_a^b \int_{\theta_{\alpha_1}}^b \cdots \int_{\theta_{\alpha_{d-1}}}^b \mathbf{K}_\alpha(\theta_1, \dots, \theta_d) \mathbf{w}(\theta_1, \dots, \theta_d) d\theta_{\alpha_d} \cdots d\theta_{\alpha_2} d\theta_{\alpha_1}, \quad \forall \mathbf{w} \in L_2[[a, b]^d] \end{aligned}$$

where $A_d \subseteq \{1, \dots, d\}^d$ denotes the set of all possible permutations of the vector $[1 \ \cdots \ d]$ and where we define the indicator function $\mathbb{I}_1(\theta_1) := 1$ if $d = 1$ or

$$\mathbb{I}_\alpha(\theta) := \mathbf{I}(\theta_{\alpha_d} - \theta_{\alpha_{d-1}}) \cdots \mathbf{I}(\theta_{\alpha_2} - \theta_{\alpha_1}), \quad \text{for} \quad \mathbf{I}(x) = \begin{cases} 1, & x \geq 0, \\ 0, & \text{else.} \end{cases}$$

¹formally, this definition deviates from the standard notion of tensor products of operators, and is really more akin to the face-splitting product

Note that the number of elements of the set A_d is $\binom{d}{d} = d!$. Also note that this class essentially incorporates all integral operators with polynomial semi-separable kernels in d variables, in the following sense.

Lemma 4. For any integrable function $\mathbf{K} : [a, b]^d \rightarrow \mathbb{R}$, we have

$$\int_a^b \cdots \int_a^b \mathbf{K}(\theta_1, \dots, \theta_d) d\theta_d \cdots d\theta_1 = \sum_{\alpha \in A_d} \int_a^b \cdots \int_a^b \mathbb{I}_\alpha(\theta_1, \dots, \theta_d) \mathbf{K}(\theta_1, \dots, \theta_d) d\theta_d \cdots d\theta_1.$$

Proof. The result follows by induction. For the base case, $d = 1$, we have $\mathbb{I}(\theta) := 1$, and the result holds trivially. Suppose for induction that the relation holds for some $d \in \mathbb{N}$. Then, for $d + 1$, we find

$$\begin{aligned} & \int_a^b \cdots \int_a^b \int_a^b \mathbf{K}(\theta_1, \dots, \theta_d, \theta_{d+1}) d\theta_{d+1} d\theta_d \cdots d\theta_1 \\ &= \sum_{\alpha \in A_d} \int_a^b \cdots \int_a^b \int_a^b \mathbb{I}_\alpha(\theta_1, \dots, \theta_d) \mathbf{K}(\theta_1, \dots, \theta_d, \theta_{d+1}) d\theta_{d+1} d\theta_d \cdots d\theta_1 \\ &= \sum_{\alpha \in A_d} \int_a^b \cdots \int_a^b \mathbb{I}_\alpha(\theta_1, \dots, \theta_d) \left[\int_a^{\theta_{\alpha_1}} \mathbf{K}(\theta_1, \dots, \theta_d, \theta_{d+1}) d\theta_{d+1} \right. \\ & \quad \left. + \sum_{j=1}^{d-1} \int_{\theta_{\alpha_j}}^{\theta_{\alpha_{j+1}}} \mathbf{K}(\theta_1, \dots, \theta_d, \theta_{d+1}) d\theta_{d+1} \right. \\ & \quad \left. + \int_{\theta_{\alpha_d}}^b \mathbf{K}(\theta_1, \dots, \theta_d, \theta_{d+1}) d\theta_{d+1} \right] d\theta_d \cdots d\theta_1 \\ &= \sum_{\alpha' \in A_{d+1}} \int_a^b \cdots \int_a^b \int_a^b \mathbb{I}_{\alpha'}(\theta_1, \dots, \theta_d, \theta_{d+1}) \mathbf{K}(\theta_1, \dots, \theta_d, \theta_{d+1}) d\theta_{d+1} d\theta_d \cdots d\theta_1. \end{aligned}$$

□

To represent this class of operators, we can use the `intop` structure, with the following fields:

- **params:** $m \times qn$ `dpvar` or `polynomial` array specifying the kernels $\mathbf{K}_\alpha \in L_2^{m \times n}[a, b]$. In particular, for $i \in \{1, \dots, q\}$, the elements `params(:, (i-1)*n+1:i*n)` specify the polynomial kernel in the i th term of the functional;
- **omat:** $q \times d$ array of integers, with each row specifying the value of α in the corresponding term in the functional. In particular, if `omat(i, :) =  $\alpha$` , this implies a term of the form

$$\int_a^b \int_{\theta_{\alpha_1}}^b \cdots \int_{\theta_{\alpha_{d-1}}}^b \mathbf{K}_i(\theta_1, \dots, \theta_d) \mathbf{w}(\theta_1, \dots, \theta_d) d\theta_{\alpha_d} \cdots d\theta_2 d\theta_1,$$

where $\mathbf{K}_i = \mathbf{K}.\text{params}(:, (i-1)*n+1:i*n)$. The number of rows q should not exceed $d!$;

- **pvarname:** $1 \times d$ `cellstr` array, specifying the names of the variables $(\theta_1, \dots, \theta_d)$. Note that the order is important, since the limits of the integral in term i are determined by `pvarname(omat(i, 1)) ≤ ... ≤ pvarname(omat(i, d))`;
- **dom:** a 1×2 array specifying the interval $[a, b]$ over which to integrate;
- **matdim:** a 1×2 array specifying the dimensions $m \times n$ of the functional.

We note that, depending on the involved monomials, the same distributed polynomial may be represented by different `intop` objects. For example, note that for any $\mathbf{x} \in L_2[a, b]$, if we have a functional on a quadratic monomial $\mathbf{x}^{\otimes 2}$,

$$\int_a^b \int_a^{\theta_1} \mathbf{K}(\theta_1, \theta_2) \mathbf{x}(\theta_1) \mathbf{x}(\theta_2) d\theta_2 d\theta_1 = \int_a^b \int_a^{\theta_2} \mathbf{K}(\theta_2, \theta_1) \mathbf{x}(\theta_2) \mathbf{x}(\theta_1) d\theta_1 d\theta_2 = \int_a^b \int_{\theta_1}^b \mathbf{K}(\theta_2, \theta_1) \mathbf{x}(\theta_1) \mathbf{x}(\theta_2) d\theta_2 d\theta_1,$$

we can represent this functional by an `intop` object $\mathbf{K}$ with `K.params =  $\mathbf{K}(\theta_1, \theta_2)$` and `K.omat =  $\begin{bmatrix} 2 & 1 \end{bmatrix}$` (indicating $\theta_2 \leq \theta_1$, or with `K.params =  $\mathbf{K}(\theta_2, \theta_1)$` and `K.omat =  $\begin{bmatrix} 1 & 2 \end{bmatrix}$` (indicating $\theta_1 \leq \theta_2$. The function `combine.terms` accounts for this non-uniqueness in the representation, combining terms that can be represented using the same integral by performing a suitable switching of the variables. Note that such a variable switching cannot be performed without changing the monomial when acting on e.g. the quadratic monomial $\mathbf{x} \otimes \mathbf{y}$ for $\mathbf{x}, \mathbf{y} \in L_2[a, b]$, which the `combine.terms` routine accounts for.

2 Operations in the polyopvar Structure

2.1 Addition in the polyopvar Representation

Consider now two distributed polynomials, $\mathbf{p}_1(\mathbf{x}) = \mathcal{C}_1 \mathbf{Z}_1(\mathbf{x})$, and $\mathbf{p}_2(\mathbf{y}) = \mathcal{C}_2 \mathbf{Z}_2(\mathbf{y})$, where the variables $\mathbf{x}, \mathbf{y}$ may or may not be the same, and where the vectors $\mathbf{Z}_1(\mathbf{x})$ and $\mathbf{Z}_2(\mathbf{y})$ may or may not include the same monomials in these variables. In order to perform addition, we simply remark that

$$\mathbf{p}_1(\mathbf{x}) + \mathbf{p}_2(\mathbf{y}) = [\mathcal{C}_1 \quad \mathcal{C}_2] \begin{bmatrix} \mathbf{Z}_1(\mathbf{x}) \\ \mathbf{Z}_2(\mathbf{y}) \end{bmatrix}$$

Here, we can combine the two monomial bases $\mathbf{Z}_1$ and $\mathbf{Z}_2$ into a unique basis $\mathbf{Z}$, establish a matrix P such that $\begin{bmatrix} \mathbf{Z}_1(x) \\ \mathbf{Z}_2(s) \end{bmatrix} = P\mathbf{Z}(x, y)$. Then the sum becomes

$$\mathbf{p}_1(\mathbf{x}) + \mathbf{p}_2(\mathbf{y}) = [\mathcal{C}_1 \quad \mathcal{C}_2] \begin{bmatrix} \mathbf{Z}_1(\mathbf{x}) \\ \mathbf{Z}_2(\mathbf{y}) \end{bmatrix} = [\mathcal{C}_1 \quad \mathcal{C}_2] P\mathbf{Z}(x, y) = \mathcal{C}\mathbf{Z}(x, y).$$

where now $\mathcal{C} = [\mathcal{C}_1 \quad \mathcal{C}_2] P$. This routine of combining the monomial bases and computing the matrix P has been implemented in the function `common_basis`. Using this function, the addition operation of `polyopvar` objects has been implemented in `plus`.

Example: For the linear polynomial $(\mathbf{p}_1(\mathbf{x}))(s_1) = \mathbf{x}(s_1)$ for $\mathbf{x} \in L_2[0, 1]$ and $\mathbf{y} \in L_2[[0, 1] \times [-1, 1]]$, we can compute the sum of $\mathbf{p}$ with itself as

```
pvar s1 s2
x = polyopvar('x', s1, [0, 1]);
p1 = x+x;
```

returning

- `p1.degmat=1` and `p2.degmat=1`
- `p1.pvarname={'s1'}`
- `p1.dom=[0, 1]`
- `p1.varmat=1`
- `p1.C={Cop}` where now `Cop` is a `nopvar` structure representing the 3-PI operator $\mathcal{C}$ given by $(\mathcal{C}\mathbf{x})(s_1) = 2\mathbf{x}(s_1)$.

Remark: Addition of e.g. $\mathbf{x} \in L_2[0, 1]$ and $\mathbf{y} \in L_2[[0, 1] \times [-1, 1]]$ should also be possible, by augmenting the operator $\mathcal{C} : \mathbf{x} \mapsto \mathbf{x}$ to an identity operator $\mathcal{C} : L_2[0, 1] \rightarrow L_2[[0, 1] \times [-1, 1]]$. However, the `nopvar` class does not currently support operators mapping between different function spaces...

2.2 Multiplication in the polyopvar Representation

Consider two distributed polynomials, $\mathbf{p}_a(\mathbf{x}) = (\mathcal{A}\mathbf{Z}_a)(\mathbf{x})(s)$, and $\mathbf{p}_b(\mathbf{y}) = (\mathcal{B}\mathbf{Z}_b)(\mathbf{y})(\theta)$, where the spatial dimensions s, θ may or may not be the same, the variables $\mathbf{x}, \mathbf{y}$ may or may not be the same, and where the vectors $\mathbf{Z}_a(\mathbf{x})$ and $\mathbf{Z}_b(\mathbf{y})$ may or may not include the same monomials in these variables. In order to perform multiplication, we start by combining the two monomial bases into a unique basis

$$\begin{bmatrix} \mathbf{Z}_a(\mathbf{x})(s) \\ \mathbf{Z}_b(\mathbf{y})(\theta) \end{bmatrix} = \begin{bmatrix} P_a \\ P_b \end{bmatrix} \mathbf{Z}(x, y)$$

where $P_a \in \mathbb{R}^{n \times \hat{n}}$, $P_b \in \mathbb{R}^{\bar{n} \times \hat{n}}$ and $\hat{n} \in \{\max(n, \bar{n}), \dots, n + \bar{n}\}$. Noting that $\tilde{\mathcal{A}}, \tilde{\mathcal{B}}$ are the tensorvar objects which equivalently represent the polynomials in the common basis, we then expand the product

$$\begin{aligned}
(\mathbf{p}_a(\mathbf{x}))(s)(\mathbf{p}_b(\mathbf{y}))(\theta) &= [\tilde{\mathcal{A}}\mathbf{Z}(\mathbf{x}, \mathbf{y})][\tilde{\mathcal{B}}\mathbf{Z}(\mathbf{x}, \mathbf{y})] \\
&= \left[\sum_{i=1}^{\hat{n}} \tilde{\mathcal{A}}_i \mathbf{Z}_i(\mathbf{x}, \mathbf{y})(s, \theta) \right] \left[\sum_{j=1}^{\hat{n}} \tilde{\mathcal{B}}_j \mathbf{Z}_j(\mathbf{x}, \mathbf{y})(s, \theta) \right] \\
&= \left[\sum_{i=1}^{\hat{n}} \sum_{k=1}^m \tilde{\mathcal{A}}_{ik} \mathbf{Z}_i(\mathbf{x}, \mathbf{y})(s, \theta) \right] \left[\sum_{j=1}^{\hat{n}} \sum_{l=1}^{\bar{m}} \tilde{\mathcal{B}}_{jl} \mathbf{Z}_j(\mathbf{x}, \mathbf{y})(s, \theta) \right] \\
&= \sum_{i=1}^{\hat{n}} \sum_{k=1}^m \sum_{j=1}^{\hat{n}} \sum_{l=1}^{\bar{m}} \left[\tilde{\mathcal{A}}_{ik} \mathbf{Z}_i(\mathbf{x}, \mathbf{y})(s, \theta) \right] \left[\tilde{\mathcal{B}}_{jl} \mathbf{Z}_j(\mathbf{x}, \mathbf{y})(s, \theta) \right]
\end{aligned}$$

If we now define $\mathbf{z} := \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}$, the above expression can be expanded as follows

$$\begin{aligned}
&= \sum_{i=1}^{\hat{n}} \sum_{k=1}^m \sum_{j=1}^{\hat{n}} \sum_{l=1}^{\bar{m}} \left[(\tilde{\mathcal{A}}_{ik,1} \mathbf{z}_1)(s, \theta) \cdots (\tilde{\mathcal{A}}_{ik,d_{i,1}} \mathbf{z}_1)(s, \theta) \cdots (\tilde{\mathcal{A}}_{ik,d_{i,1}+\cdots+d_{i,p-1}+1} \mathbf{z}_p)(s, \theta) \cdots (\tilde{\mathcal{A}}_{ik,d_{i,1}+\cdots+d_{i,p}} \mathbf{z}_p)(s, \theta) \right] \\
&\quad \left[(\tilde{\mathcal{B}}_{jl,1} \mathbf{z}_1)(s, \theta) \cdots (\tilde{\mathcal{B}}_{jl,d_{j,1}} \mathbf{z}_1)(s, \theta) \cdots (\tilde{\mathcal{B}}_{jl,d_{j,1}+\cdots+d_{j,p-1}+1} \mathbf{z}_p)(s, \theta) \cdots (\tilde{\mathcal{B}}_{jl,d_{j,1}+\cdots+d_{j,p}} \mathbf{z}_p)(s, \theta) \right] \\
&= \sum_{i=1}^{\hat{n}} \sum_{k=1}^m \sum_{j=1}^{\hat{n}} \sum_{l=1}^{\bar{m}} (\tilde{\mathcal{A}}_{ik,1} \mathbf{z}_1)(s, \theta) \cdots (\tilde{\mathcal{A}}_{ik,d_{i,1}} \mathbf{z}_1)(s, \theta) (\tilde{\mathcal{B}}_{jl,1} \mathbf{z}_1)(s, \theta) \cdots (\tilde{\mathcal{B}}_{jl,d_{j,1}} \mathbf{z}_1)(s, \theta) \cdots \\
&\quad (\tilde{\mathcal{A}}_{ik,d_{i,1}+\cdots+d_{i,p-1}+1} \mathbf{z}_p)(s, \theta) \cdots (\tilde{\mathcal{A}}_{ik,d_{i,1}+\cdots+d_{i,p}} \mathbf{z}_p)(s, \theta) \cdots \\
&\quad (\tilde{\mathcal{B}}_{jl,d_{j,1}+\cdots+d_{j,p-1}+1} \mathbf{z}_p)(s, \theta) \cdots (\tilde{\mathcal{B}}_{jl,d_{j,1}+\cdots+d_{j,p}} \mathbf{z}_p)(s, \theta)
\end{aligned}$$

where the new monomial basis is given by

$$\mathbf{Z}(z)(s, \theta) \otimes \mathbf{Z}(z)(s, \theta) = \begin{bmatrix} z_1^{\otimes d_{1,1}+d_{1,1}} \otimes \cdots \otimes z_p^{\otimes d_{1,p}+d_{1,p}} \\ \vdots \\ z_1^{\otimes d_{1,1}+d_{\hat{n},1}} \otimes \cdots \otimes z_p^{\otimes d_{1,p}+d_{\hat{n},p}} \\ \vdots \\ z_1^{\otimes d_{\hat{n},1}+d_{1,1}} \otimes \cdots \otimes z_p^{\otimes d_{\hat{n},p}+d_{1,p}} \\ \vdots \\ z_1^{\otimes d_{\hat{n},1}+d_{\hat{n},1}} \otimes \cdots \otimes z_p^{\otimes d_{\hat{n},p}+d_{\hat{n},p}} \end{bmatrix}$$

and

3 Lyapunov Stability Analysis of Polynomial PIEs

We now propose a procedure for testing stability of the origin for a distributed polynomial PIE, by finding a distributed polynomial Lyapunov functional that decreases along solutions to the PIE. The procedure for finding such a Lyapunov functional consists of three key steps:

1. Parameterize distributed SOS functionals V and W ;
2. Compute the derivative $\dot{V}$ of V along the PIE;
3. Enforce $\dot{V} = -W$;

In the following subsections, we show how each of these steps can be performed in the PIESOS framework.

3.1 Parameterizing Distributed SOS Functionals: `piesos_sosvar`

We first consider the challenge of parameterizing distributed SOS functionals. This procedure has been implemented in the function `piesos_sosvar`, which takes as input a basis of distributed monomials $\mathbf{Z}_d(\mathbf{x})$, and returns a degree- $2d$ distributed polynomial functional $V(\mathbf{x})$ that is nonnegative for any value of $\mathbf{x}$. This functional is generated in two steps:

First, parameterize V in the quadratic form as

$$V(\mathbf{x}) = \langle \mathbf{Z}_d(\mathbf{x}), \mathcal{P} \mathbf{Z}_d(\mathbf{x}) \rangle_{L_2} = \left\langle \begin{bmatrix} \mathbf{x} \\ \vdots \\ \mathbf{x}^{\otimes d} \end{bmatrix}, \begin{bmatrix} \mathcal{P}_{11} & \cdots & \mathcal{P}_{1d} \\ \vdots & \ddots & \vdots \\ \mathcal{P}_{d1} & \cdots & \mathcal{P}_{dd} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \vdots \\ \mathbf{x}^{\otimes d} \end{bmatrix} \right\rangle_{L_2},$$

for $\mathbf{x} \in L_2[a, b]$, and where $\mathcal{P}_{ij} : L_2[[a, b]^j] \rightarrow L_2[[a, b]^i]$ are such that $\mathcal{P} \succeq 0$ (and thus $V(\mathbf{x}) \geq 0$). Here, the operator $\mathcal{P} \succeq 0$ is generated with the function `piesos_poslpivar`.

Next, convert V to the linear form,

$$V(\mathbf{x}) = \text{vec}(\mathcal{P}) \mathbf{Z}_{2d}(\mathbf{x}). \quad (4)$$

The operator $\mathcal{K} := \text{vec}(\mathcal{P})$ is computed using the function `quad2lin`.

We show how each of these steps is performed in the following subsections.

3.1.1 Parameterizing Positive Operators: `piesos_poslpivar`

Suppose we want to parameterize a positive semidefinite operator $\mathcal{P}$ acting on the degree d distributed monomial basis, Lyapunov functional

$$\mathcal{P} \mathbf{Z}_d(x) = \begin{bmatrix} \mathcal{P}_{11} & \cdots & \mathcal{P}_{1d} \\ \vdots & \ddots & \vdots \\ \mathcal{P}_{d1} & \cdots & \mathcal{P}_{dd} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \vdots \\ \mathbf{x}^{\otimes d} \end{bmatrix},$$

for $\mathbf{x} \in L_2[a, b]$ and $\mathcal{P}_{ij} : L_2[[a, b]^j] \rightarrow L_2[[a, b]^i]$. Here, $\mathcal{P}_{11}$ will be just a 3-PI operator, so we can parameterize $\mathcal{P}_{11}$ using the basis of 3-PI operators,

$$(\mathcal{Z} \mathbf{x})(s) := \begin{bmatrix} (\mathcal{Z}_0 \mathbf{x})(s) \\ (\mathcal{Z}_1 \mathbf{x})(s) \\ (\mathcal{Z}_2 \mathbf{x})(s) \end{bmatrix} := \begin{bmatrix} \mathbf{Z}_d(s) \mathbf{x}(s) \\ \int_a^s \mathbf{Z}_d(s, \theta) \mathbf{x}(\theta) d\theta \\ \int_s^b \mathbf{Z}_d(s, \theta) \mathbf{x}(\theta) d\theta \end{bmatrix},$$

where $\mathbf{Z}_d(s, \theta)$ is the basis of monomials of degree at most d in $s, \theta \in [a, b]$. Then, we can parameterize the operator $\mathcal{P}_{11} \succeq 0$ by a matrix $P_{11} \succeq 0$ as

$$\mathcal{P}_{11} = \mathcal{Z}^* P_{11} \mathcal{Z} = \begin{bmatrix} \mathcal{Z}_0^* & \mathcal{Z}_1^* & \mathcal{Z}_2^* \end{bmatrix} P_{11} \begin{bmatrix} \mathcal{Z}_0 \\ \mathcal{Z}_1 \\ \mathcal{Z}_2 \end{bmatrix}.$$

Similarly, to parameterize $\mathcal{P}_{12} : L_2[[a, b]^2] \rightarrow L_2[a, b]$, consider the tensor product of the basis operators, $\mathcal{Z}_i \otimes \mathcal{Z}_j : L_2[[a, b]^2] \rightarrow L_2[a, b]$, defined by

$$(\mathcal{Z}_i \otimes \mathcal{Z}_j \mathbf{x}^{\otimes 2})(s) = (\mathcal{Z}_i \mathbf{x})(s) (\mathcal{Z}_j \mathbf{x})(s),$$

for $i, j \in \{1, 2\}$. Then we can parameterize $\mathcal{P}_{12}$ by a matrix P_{12} as

$$\mathcal{P}_{12} = \mathcal{Z}^* P_{12} [\hat{\mathcal{Z}} \otimes \hat{\mathcal{Z}}] = \begin{bmatrix} \mathcal{Z}_0^* & \mathcal{Z}_1^* & \mathcal{Z}_2^* \end{bmatrix} P_{12} \begin{bmatrix} \mathcal{Z}_1 \otimes \mathcal{Z}_1 \\ \mathcal{Z}_1 \otimes \mathcal{Z}_2 \\ \mathcal{Z}_2 \otimes \mathcal{Z}_1 \\ \mathcal{Z}_2 \otimes \mathcal{Z}_2 \end{bmatrix},$$

where we define $\hat{\mathcal{Z}} := \begin{bmatrix} \mathcal{Z}_1 \\ \mathcal{Z}_2 \end{bmatrix}$. Similarly, we can parameterize $\mathcal{P}_{22}$ by a matrix P_{22} as

$$\mathcal{P}_{22} = [\hat{\mathcal{Z}} \otimes \hat{\mathcal{Z}}]^* P_{22} [\hat{\mathcal{Z}} \otimes \hat{\mathcal{Z}}] = \begin{bmatrix} (\mathcal{Z}_1 \otimes \mathcal{Z}_1)^* & (\mathcal{Z}_1 \otimes \mathcal{Z}_2)^* & (\mathcal{Z}_2 \otimes \mathcal{Z}_1)^* & (\mathcal{Z}_2 \otimes \mathcal{Z}_2)^* \end{bmatrix} P_{22} \begin{bmatrix} \mathcal{Z}_1 \otimes \mathcal{Z}_1 \\ \mathcal{Z}_1 \otimes \mathcal{Z}_2 \\ \mathcal{Z}_2 \otimes \mathcal{Z}_1 \\ \mathcal{Z}_2 \otimes \mathcal{Z}_2 \end{bmatrix}.$$

Note here that $(\mathcal{Z}_i \otimes \mathcal{Z}_j)^* \neq \mathcal{Z}_i^* \otimes \mathcal{Z}_j^*$, since $(\mathcal{Z}_i \otimes \mathcal{Z}_j)^* : L_2[a, b] \rightarrow L_2[[a, b]^2]$. More generally, we parameterize

$$\mathcal{P} = \begin{bmatrix} \mathcal{P}_{11} & \mathcal{P}_{12} & \cdots & \mathcal{P}_{1d} \\ \mathcal{P}_{21} & \mathcal{P}_{22} & \cdots & \mathcal{P}_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \mathcal{P}_{d1} & \mathcal{P}_{d2} & \cdots & \mathcal{P}_{dd} \end{bmatrix} = \begin{bmatrix} \mathcal{Z}^* & & & \\ & (\hat{\mathcal{Z}} \otimes \hat{\mathcal{Z}})^* & & \\ & & \ddots & \\ & & & (\hat{\mathcal{Z}}^{\otimes d})^* \end{bmatrix} \begin{bmatrix} \mathcal{P}_{11} & \mathcal{P}_{12} & \cdots & \mathcal{P}_{1d} \\ \mathcal{P}_{21} & \mathcal{P}_{22} & \cdots & \mathcal{P}_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \mathcal{P}_{d1} & \mathcal{P}_{d2} & \cdots & \mathcal{P}_{dd} \end{bmatrix} \begin{bmatrix} \mathcal{Z} & & & \\ & (\hat{\mathcal{Z}} \otimes \hat{\mathcal{Z}}) & & \\ & & \ddots & \\ & & & \hat{\mathcal{Z}}^{\otimes d} \end{bmatrix}.$$

A routine for declaring such an operator $\mathcal{P}$ has been implemented in `piesos_poslpivar`. This routine takes a distributed monomial basis, e.g. $\begin{bmatrix} \mathbf{x} \\ \vdots \\ \mathbf{x}^{\otimes d} \end{bmatrix}$ or $\begin{bmatrix} \mathbf{x}_1^{\otimes \mathbf{x}_2} \\ \vdots \\ \mathbf{x}_1^{\otimes \mathbf{x}_d} \end{bmatrix}$, and returns a basis operator $\mathcal{Z}$ acting on this distributed monomial basis, as well as a positive semidefinite matrix variable P .

3.1.2 Converting to Linear Format: quad2lin

Consider now a distributed polynomial functional in quadratic form,

$$V(\mathbf{x}) = \langle \mathbf{Z}_d(\mathbf{x}), \mathcal{P} \mathbf{Z}_d(\mathbf{x}) \rangle_{L_2} = \left\langle \begin{bmatrix} \mathbf{x} \\ \vdots \\ \mathbf{x}^{\otimes d} \end{bmatrix}, \begin{bmatrix} \mathcal{P}_{11} & \cdots & \mathcal{P}_{1d} \\ \vdots & \ddots & \vdots \\ \mathcal{P}_{d1} & \cdots & \mathcal{P}_{dd} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \vdots \\ \mathbf{x}^{\otimes d} \end{bmatrix} \right\rangle_{L_2} = \sum_{i=1}^d \sum_{j=1}^d \langle \mathbf{x}^{\otimes i}, \mathcal{P}_{ij} \mathbf{x}^{\otimes j} \rangle_{L_2}.$$

Parameterizing the operators $\mathcal{P}_{ij}$ as before, we get

$$\langle \mathbf{x}^{\otimes i}, \mathcal{P}_{ij} \mathbf{x}^{\otimes j} \rangle_{L_2} = \left\langle \hat{\mathcal{Z}}^{\otimes i} \mathbf{x}^{\otimes i}, \mathcal{P}_{ij} \hat{\mathcal{Z}}^{\otimes j} \mathbf{x}^{\otimes j} \right\rangle_{L_2} = \int_a^b \overbrace{[(\hat{\mathcal{Z}} \mathbf{x})(s)^T \otimes \cdots \otimes (\hat{\mathcal{Z}} \mathbf{x})(s)^T]}^{i \text{ products}} P_{ij} \overbrace{[(\hat{\mathcal{Z}} \mathbf{x})(s) \otimes \cdots \otimes (\hat{\mathcal{Z}} \mathbf{x})(s)]}^{j \text{ products}} ds$$

for each $i, j \in \{1, \dots, d\}$, where in this case $\otimes$ denotes just the Kronecker product. **I wonder if we need to generalise the Tensor-PI operators (or define a new class) to Kronecker-PI operators for terms such as $\hat{\mathcal{Z}}^{\otimes i} \mathbf{x}^{\otimes i}$?** To evaluate this expression, we need to consider integrals of the form

$$\int_a^b (\mathcal{R}_1 \mathbf{x})(s) \cdots (\mathcal{R}_n \mathbf{x})(s) ds$$

for 2-PI operators $\mathcal{R}_i$. **But P_{ij} is a 3-PI operator? Also the integrand from the inner product is not in the form of a T-PI operator.** For this, we define the following “vectorization” of tensor-PI operators.

Definition 5. For given polynomial parameters $\mathbf{R}_{i,1}, \mathbf{R}_{i,2} \in \mathbb{R}[s, \theta_i]$ for $i \in \{1, \dots, d\}$, define associated functions $\mathbf{K}_\alpha \in \mathbb{R}[\theta_1, \dots, \theta_d]$ for each $\alpha \in A_d$ as

$$\mathbf{K}_\alpha(\theta) = \sum_{j=0}^d \int_{\theta_{\alpha_j}}^{\theta_{\alpha_{j+1}}} \mathbf{R}_{\alpha_1, \gamma(1,j)}(s, \theta_{\alpha_1}) \cdots \mathbf{R}_{\alpha_d, \gamma(d,j)}(s, \theta_{\alpha_d}) ds,$$

where we let $\theta_{\alpha_0} := a$ and $\theta_{\alpha_{d+1}} := b$, and where

$$\gamma(k, j) := \begin{cases} 1, & k \leq j, \\ 2, & j > k, \end{cases} \quad k, j \in \{1, \dots, d\}.$$

For a given tensor product of 2-PI operators, $\mathcal{P} := \mathcal{R}_1 \otimes \cdots \otimes \mathcal{R}_d : L_2[[a, b]^d] \rightarrow L_2[a, b]$, where each $\mathcal{R}_i : L_2[a, b] \rightarrow L_2[a, b]$ is defined by

$$(\mathcal{R}_i \mathbf{v})(s) = \int_a^s \mathbf{R}_{i,1}(s, \theta) \mathbf{v}(\theta) d\theta + \int_s^b \mathbf{R}_{i,2}(s, \theta) \mathbf{v}(\theta) d\theta$$

we then define $\text{vec}(\mathcal{P}) := \mathcal{K} : L_2[[a, b]^d] \rightarrow \mathbb{R}$ by

$$\mathcal{K} \mathbf{w} := \sum_{\alpha \in A_d} \int_a^b \int_{\theta_{\alpha_1}}^b \cdots \int_{\theta_{\alpha_{d-1}}}^b \mathbf{K}_{\alpha}(\theta_1, \dots, \theta_d) \mathbf{w}(\theta_1, \dots, \theta_d) d\theta_{\alpha_d} \cdots d\theta_{\alpha_2} d\theta_{\alpha_1}.$$

For a given tensor-PI operator, $\mathcal{P} := \sum_{i=1}^m \mathcal{P}_i = \sum_{i=1}^m \mathcal{R}_{i,1} \otimes \cdots \otimes \mathcal{R}_{i,d} : L_2[[a, b]^d] \rightarrow L_2[a, b]$, we define $\text{vec}(\mathcal{P}) := \sum_{j=1}^m \text{vec}(\mathcal{P}_i)$.

The following lemma shows that the vectorization of a tensor-PI operator corresponds to the composition of a functional with this operator.

Lemma 6. For any 1D tensor-PI operator, $\mathcal{P} : L_2[[a, b]^d] \rightarrow L_2[a, b]$, we have

$$\int_a^b (\mathcal{P} \mathbf{w})(s) ds = \text{vec}(\mathcal{P}) \mathbf{w}, \quad \forall \mathbf{w} \in L_2[[a, b]^d].$$

Proof. Let parameters $\mathbf{R}_{i,1}, \mathbf{R}_{i,2} \in \mathbb{R}[s, \theta_i]$ for $i \in \{1, \dots, d\}$ be such that $\mathcal{P} := \mathcal{R}_1 \otimes \cdots \otimes \mathcal{R}_d : L_2[[a, b]^d] \rightarrow L_2[a, b]$, with

$$(\mathcal{R}_i \mathbf{v})(s) = \int_a^s \mathbf{R}_{i,1}(s, \theta) \mathbf{v}(\theta) d\theta + \int_s^b \mathbf{R}_{i,2}(s, \theta) \mathbf{v}(\theta) d\theta. \quad (5)$$

Then, we can express

$$\int_a^b (\mathcal{R}_1 \mathbf{x}_1)(s) \cdots (\mathcal{R}_d \mathbf{x}_d)(s) ds = \sum_{\beta \in \{1,2\}^d} \int_a^b \int_a^b \cdots \int_a^b \prod_{i=1}^d [\mathbf{I}_{\beta_i}(s, \theta_i) \mathbf{R}_{i,\beta_i}(s, \theta_i) \mathbf{x}_i(\theta_i)] d\theta_d \cdots d\theta_1 ds,$$

where we define the indicator function $\mathbf{I}_1(x, y) = \mathbf{I}_2(y, x) := \mathbf{I}(x - y)$, where $\mathbf{I}(z) := \begin{cases} 0, & z \leq 0, \\ 1, & z > 0 \end{cases}$. Defining $\mathbf{w}(\theta) := \mathbf{x}_1(\theta_1) \cdots \mathbf{x}_d(\theta_d)$, and applying Lemma 4, it follows that

$$\begin{aligned} \int_a^b \mathcal{P} \mathbf{w}(s) ds &= \int_a^b \cdots \int_a^b \left[\sum_{\beta \in \{1,2\}^d} \int_a^b \prod_{i=1}^d [\mathbf{I}_{\beta_i}(s, \theta_i) \mathbf{R}_{i,\beta_i}(s, \theta_i)] ds \right] \mathbf{w}(\theta_1, \dots, \theta_d) d\theta_d \cdots d\theta_1 \\ &= \sum_{\alpha \in A_d} \int_a^b \cdots \int_a^b \mathbb{I}_{\alpha}(\theta_1, \dots, \theta_d) \left[\sum_{\beta \in \{1,2\}^d} \int_a^b \prod_{i=1}^d [\mathbf{I}_{\beta_i}(s, \theta_i) \mathbf{R}_{i,\beta_i}(s, \theta_i)] ds \right] \mathbf{w}(\theta_1, \dots, \theta_d) d\theta_d \cdots d\theta_1, \end{aligned}$$

where $\mathbb{I}_{\alpha}(\theta) := \mathbf{I}(\theta_{\alpha_d} - \theta_{\alpha_{d-1}}) \cdots \mathbf{I}(\theta_{\alpha_2} - \theta_{\alpha_1})$. Now, consider arbitrary $\alpha \in A_d$. Then, we can decompose

$$\begin{aligned} \mathbb{I}_{\alpha}(\theta) \sum_{\beta \in \{1,2\}^d} \int_a^b \prod_{i=1}^d [\mathbf{I}_{\beta_i}(s, \theta_i) \mathbf{R}_{i,\beta_i}(s, \theta_i)] ds &= \sum_{\beta \in \{1,2\}^d} \sum_{j=0}^d \int_{\theta_{\alpha_j}}^{\theta_{\alpha_{j+1}}} \mathbb{I}_{\alpha}(\theta) \prod_{i=1}^d [\mathbf{I}_{\beta_i}(s, \theta_i) \mathbf{R}_{i,\beta_i}(s, \theta_i)] ds, \\ &= \sum_{\beta \in \{1,2\}^d} \sum_{j=0}^d \int_{\theta_{\alpha_j}}^{\theta_{\alpha_{j+1}}} \mathbb{I}_{\alpha}(\theta) \prod_{i=1}^d [\mathbf{I}_{\beta_i}(s, \theta_{\alpha_i}) \mathbf{R}_{\alpha_i, \beta_i}(s, \theta_{\alpha_i})] ds, \end{aligned}$$

where we define $\alpha_0 = 0$, $\alpha_{d+1} = d + 1$, and $\theta_0 := a$ and $\theta_{d+1} := b$. Defining, then

$$\gamma(k, j) := \begin{cases} 1, & k \leq j, \\ 2, & k > j, \end{cases} \quad k, j \in \{1, \dots, d\}.$$

it follows by definition of $\mathbf{I}_{\beta_i}(s, \theta_i)$ that

$$\mathbb{I}_{\alpha}(\theta) \sum_{\beta \in \{1,2\}^d} \int_a^b \prod_{i=1}^d [\mathbf{I}_{\beta_i}(s, \theta_i) \mathbf{R}_{i,\beta_i}(s, \theta_i)] ds = \sum_{j=0}^d \int_{\theta_{\alpha_j}}^{\theta_{\alpha_{j+1}}} \prod_{i=1}^d [\mathbf{R}_{\alpha_i, \gamma(i,j)}(s, \theta_{\alpha_i})] ds = \mathbf{K}_{\alpha}(\theta)$$

whence

$$\int_a^b (\mathcal{P} \mathbf{w})(s) ds = \text{vec}(\mathcal{P}) \mathbf{w}, \quad \forall \mathbf{w} \in L_2[[a, b]^d].$$

□

Using the vectorization operation, we can also convert from the quadratic to the linear format. In particular, we define the following vectorization of a quadratic form.

Definition 7. For given $\mathcal{P} : L_2[[a, b]^k] \rightarrow L_2[[a, b]^\ell]$ of the form

$$\mathcal{P} = (\mathcal{Z}_{L,1} \otimes \cdots \otimes \mathcal{Z}_{L,k})^* P (\mathcal{Z}_{R,1} \otimes \cdots \otimes \mathcal{Z}_{R,\ell})$$

where $\mathcal{Z}_{L,i} : L_2[a, b] \rightarrow L_2^m[a, b]$ and $\mathcal{Z}_{R,i} : L_2[a, b] \rightarrow L_2^n[a, b]$ are 2-PI operators and $P \in \mathbb{R}^{m \times n}$, we define

$$\text{vec}(\mathcal{P}) := \sum_{i=1}^m \sum_{j=1}^n P_{ij} \text{vec}([\mathcal{Z}_{L,1} \otimes \cdots \otimes \mathcal{Z}_{L,k}]_i \otimes [\mathcal{Z}_{R,1} \otimes \cdots \otimes \mathcal{Z}_{R,\ell}]_j)$$

Lemma 8. For any $\mathcal{P} : L_2[[a, b]^k] \rightarrow L_2[[a, b]^\ell]$ of the form

$$\mathcal{P} = (\mathcal{Z}_{L,1} \otimes \cdots \otimes \mathcal{Z}_{L,k})^* P (\mathcal{Z}_{R,1} \otimes \cdots \otimes \mathcal{Z}_{R,\ell}),$$

we have

$$\langle \mathbf{w}, \mathcal{P}\mathbf{u} \rangle_{L_2} = \text{vec}(\mathcal{P})(\mathbf{w} \otimes \mathbf{u}), \quad \forall \mathbf{w} \in L_2[[a, b]^k], \mathbf{u} \in L_2[[a, b]^\ell].$$

Proof. For any $\mathbf{w} \in L_2[[a, b]^k]$ and $\mathbf{u} \in L_2[[a, b]^\ell]$, we have by Lem. 6

$$\begin{aligned} \langle \mathbf{w}, \mathcal{P}\mathbf{u} \rangle_{L_2} &= \langle (\mathcal{Z}_{L,1} \otimes \cdots \otimes \mathcal{Z}_{L,k})\mathbf{w}, P(\mathcal{Z}_{R,1} \otimes \cdots \otimes \mathcal{Z}_{R,\ell})\mathbf{u} \rangle_{L_2} \\ &= \sum_{i=1}^m \sum_{j=1}^n \int_a^b ((\mathcal{Z}_{L,1} \otimes \cdots \otimes \mathcal{Z}_{L,k})\mathbf{w})(s) P_{ij} ((\mathcal{Z}_{R,1} \otimes \cdots \otimes \mathcal{Z}_{R,\ell})\mathbf{u})(s) ds \\ &= \sum_{i=1}^m \sum_{j=1}^n P_{ij} \int_a^b ((\mathcal{Z}_{L,1} \otimes \cdots \otimes \mathcal{Z}_{L,k} \otimes \mathcal{Z}_{R,1} \otimes \cdots \otimes \mathcal{Z}_{R,\ell})(\mathbf{w} \otimes \mathbf{u}))(s) ds = \text{vec}(\mathcal{P})(\mathbf{w} \otimes \mathbf{u}). \end{aligned}$$

□

A routine computing this vectorization for given P , $\mathcal{Z}_L$, and $\mathcal{Z}_R$, has been implemented in `quad2lin.term`. Using this function, we can also convert the quadratic form

$$V(\mathbf{x}) = \langle \mathbf{Z}_d(\mathbf{x}), \mathcal{P}\mathbf{Z}_d(\mathbf{x}) \rangle_{L_2} = \sum_{i=1}^d \sum_{j=1}^d \langle \mathbf{x}^{\otimes i}, \mathcal{P}_{ij} \mathbf{x}^{\otimes j} \rangle_{L_2} \quad (6)$$

to the linear form, which has been implemented in the function `quad2lin`.

3.2 Computing the Lie Derivative of the Lyapunov Functional: Liediff

Having established how we can parameterize a distributed SOS polynomial function V , we now consider the challenge of taking the Lie derivative of this functional along a given PIE.

Suppose we have a polynomial PIE

$$\mathcal{T}\mathbf{x}_t(t) = \mathbf{f}(\mathbf{x}(t)) = \mathcal{C}_1\mathbf{x}(t) + \cdots + \mathcal{C}_d\mathbf{x}(t)^{\otimes d},$$

and a distributed polynomial Lyapunov functional

$$V(\mathbf{x}) = \mathcal{K}\mathbf{Z}_q(\mathcal{T}\mathbf{x}) = \mathcal{K}_1(\mathcal{T}\mathbf{x}) + \cdots + \mathcal{K}_q(\mathcal{T}\mathbf{x})^{\otimes q}$$

Then the derivative of the Lyapunov functional along solutions to the PIE is given by

$$\dot{V}(\mathbf{x}(t)) = \sum_{j=1}^q \frac{d}{dt} \mathcal{K}_j(\mathcal{T}\mathbf{x}(t))^{\otimes j}$$

The following lemma shows how we can compute this derivative by performing suitable substitution of the variables $\mathcal{T}\mathbf{x}$ with $\mathcal{C}\mathbf{Z}_d(\mathbf{x})$.

Lemma 9. For each $k \in \{1, \dots, q\}$, let $\mathcal{K}_k : L_2[\Omega^k] \rightarrow \mathbb{R}$ be an F-PI operator for $\Omega = [a, b]$, and define the functionals

$$V(\mathbf{x}) = [\mathcal{K}_1 \quad \dots \quad \mathcal{K}_q] \mathbf{Z}_q(\mathcal{T}\mathbf{x}) = \sum_{k=1}^q \mathcal{K}_k(\mathcal{T}\mathbf{x})^{\otimes k}, \quad W_k(\mathbf{y}_1, \dots, \mathbf{y}_d) := \mathcal{K}_k[\mathbf{y}_1 \otimes \dots \otimes \mathbf{y}_d].$$

for $\mathbf{x}, \mathbf{y}_i \in \Omega$. Then, for a polynomial PIE defined by $\{\mathcal{T}, \mathcal{C}\}$, we have

$$\mathcal{L}_{\mathcal{T}, \mathcal{C}} V(\mathbf{x}) = \sum_{k=1}^q \left(W_k(\mathcal{C}\mathbf{Z}_d(\mathbf{x}), \mathcal{T}\mathbf{x}, \dots, \mathcal{T}\mathbf{x}) + W_k(\mathcal{T}\mathbf{x}, \mathcal{C}\mathbf{Z}_d(\mathbf{x}), \mathcal{T}\mathbf{x}, \dots, \mathcal{T}\mathbf{x}) + \dots + W_k(\mathcal{T}\mathbf{x}, \dots, \mathcal{T}\mathbf{x}, \mathcal{C}\mathbf{Z}_d(\mathbf{x})) \right)$$

Proof. The result follows immediately by the product rule, by which, for each $k \in \{1, \dots, q\}$ we have

$$\begin{aligned} \frac{d}{dt} (\mathcal{T}\mathbf{x}(t))^{\otimes k} &= \sum_{j=0}^k (\mathcal{T}\mathbf{x}(t))^{\otimes j} \otimes \left(\frac{d}{dt} \mathcal{T}\mathbf{x}(t) \right) \otimes (\mathcal{T}\mathbf{x}(t))^{\otimes q-j-1} \\ &= \sum_{j=0}^k (\mathcal{T}\mathbf{x}(t))^{\otimes j} \otimes \left(\mathcal{C}\mathbf{Z}_d(\mathbf{x}(t)) \right) \otimes (\mathcal{T}\mathbf{x}(t))^{\otimes d-j-1}, \end{aligned}$$

wherefore

$$\frac{d}{dt} V(\mathbf{x}(t)) = \sum_{k=1}^q \mathcal{K}_k(\mathcal{T}\mathbf{x})^{\otimes k} = \sum_{k=1}^q \left(W_k(\mathcal{C}\mathbf{Z}_d(\mathbf{x}), \mathcal{T}\mathbf{x}, \dots, \mathcal{T}\mathbf{x}) + \dots + W_k(\mathcal{T}\mathbf{x}, \dots, \mathcal{T}\mathbf{x}, \mathcal{C}\mathbf{Z}_d(\mathbf{x})) \right).$$

□

Applying this lemma requires substituting a variable in a distributed polynomial functional by another distributed polynomial. In order to perform this substitution, the following lemma first shows how we may substitute a variable in a homogeneous distributed polynomial for a distributed polynomial.

Lemma 10. For $\Omega = [a, b] \subseteq \mathbb{R}$ and $n, d \in \mathbb{N}$, let

$$p(\mathbf{x}_1, \dots, \mathbf{x}_n) = \mathcal{K}(\mathbf{x}_1 \otimes \dots \otimes \mathbf{x}_n), \quad q(\mathbf{y}) = \mathcal{P}\mathbf{Z}_d(\mathbf{y}) = \sum_{k=1}^d \mathcal{P}_k \mathbf{y}^{\otimes k},$$

for a functional PI-operator $\mathcal{K}$, and tensor-PI operators $\mathcal{P}_k = \sum_{\ell=1}^M \mathcal{P}_{k,\ell}$ where $\mathcal{P}_{k,\ell} := \mathcal{R}_{k,\ell,1} \otimes \dots \otimes \mathcal{R}_{k,\ell,d}$ for 2-PI operators $\mathcal{R}_{k,\ell,i}$. Then, for any $i \in \{1, \dots, p\}$

$$p(\mathbf{x}_1, \dots, \mathbf{x}_{i-1}, q(\mathbf{y}), \mathbf{x}_i, \dots, \mathbf{x}_{n-1}) = \sum_{k=1}^d \sum_{\ell=1}^M p(\mathbf{x}_1, \dots, \mathbf{x}_{i-1}, \mathcal{P}_{k,\ell} \mathbf{y}^{\otimes k}, \mathbf{x}_i, \dots, \mathbf{x}_{n-1}) \quad (7)$$

for any $\mathbf{x}_1, \dots, \mathbf{x}_{n-1} \in L_2[\Omega]$ and $\mathbf{y} \in L_2[\Omega^d]$.

Proof. Let $p(\mathbf{x}_1, \dots, \mathbf{x}_n) = \mathcal{K}(\mathbf{x}_1 \otimes \dots \otimes \mathbf{x}_n)$. Then, by linearity of the F-PI operator $\mathcal{K}$,

$$\begin{aligned} p(\mathbf{x}_1, \dots, \mathbf{x}_{i-1}, q(\mathbf{y}), \mathbf{x}_i, \dots, \mathbf{x}_{n-1}) &= \mathcal{K} \left(\mathbf{x}_1 \otimes \dots \otimes \mathbf{x}_{i-1} \otimes \left[\sum_{k=1}^d \sum_{\ell=1}^M \mathcal{P}_{k,\ell} \mathbf{y}^{\otimes k} \right] \otimes \mathbf{x}_i \otimes \dots \otimes \mathbf{x}_{n-1} \right) \\ &= \sum_{k=1}^d \sum_{\ell=1}^M \mathcal{K} \left(\mathbf{x}_1 \otimes \dots \otimes \mathbf{x}_{i-1} \otimes \left[\mathcal{P}_{k,\ell} \mathbf{y}^{\otimes k} \right] \otimes \mathbf{x}_i \otimes \dots \otimes \mathbf{x}_{n-1} \right) \\ &= \sum_{k=1}^d \sum_{\ell=1}^M p(\mathbf{x}_1, \dots, \mathbf{x}_{i-1}, \mathcal{P}_{k,\ell} \mathbf{y}^{\otimes k}, \mathbf{x}_i, \dots, \mathbf{x}_{n-1}). \end{aligned}$$

□

Lem. 10 shows how substitution of a single factor in a homogeneous polynomial by a distributed polynomial can be expressed as a sum of polynomials, wherein the variable is substituted for a homogeneous polynomial. In order to perform this substitution, the following lemma shows that substitution may always be performed with respect to the last factor in the homogeneous polynomial.

Lemma 11. For any $\Omega = [a, b] \subseteq \mathbb{R}$, $n \in \mathbb{N}$, and $K_\alpha \in \mathbb{R}[\mathbf{s}_n]$ for $\alpha \in A_d$, let

$$p(\mathbf{x}_1, \dots, \mathbf{x}_n) = \sum_{\alpha \in A_n} \int_{\Omega^n} \mathbb{I}_\alpha(\mathbf{s}_n) K_\alpha(\mathbf{s}_n) \mathbf{x}_1(s_1) \cdots \mathbf{x}_n(s_n) d\mathbf{s}_n.$$

Then, for any $i \in \{1, \dots, n\}$, and $\mathbf{x}_1, \dots, \mathbf{x}_{n-1}, \mathbf{y} \in L_2[\Omega]$, we have

$$p(\mathbf{x}_1, \dots, \mathbf{x}_{i-1}, \mathbf{y}, \mathbf{x}_i, \dots, \mathbf{x}_{n-1}) = \hat{p}(\mathbf{x}_1, \dots, \mathbf{x}_{n-1}, \mathbf{y}),$$

where

$$\hat{p}(\mathbf{x}_1, \dots, \mathbf{x}_{n-1}, \mathbf{y}) = \sum_{\alpha \in A_n} \int_{\Omega^n} \mathbb{I}_\alpha(\mathbf{s}_n) \hat{K}_\alpha(\mathbf{s}_n) \mathbf{x}_1(s_1) \cdots \mathbf{x}_{n-1}(s_{n-1}) \mathbf{y}(s_n) d\mathbf{s}_n,$$

where for each $\alpha \in A_d$, $\hat{K}_{\hat{\alpha}}(\hat{\mathbf{s}}_n) := K_\alpha(\mathbf{s}_n)$ for

$$\hat{s}_j := \begin{cases} s_j, & j < i, \\ s_{j+1}, & i \leq j < n, \\ s_i, & j = n, \end{cases} \quad \hat{\alpha}_k := \begin{cases} \alpha_k, & \alpha_k < i, \\ \alpha_k - 1, & i < \alpha_k \leq n, \\ n, & \alpha_k = i, \end{cases}$$

Proof. For arbitrary $\alpha \in A_n$ and $s \in \Omega^n$, let $\hat{\alpha}$ and $\hat{s}$ be as defined. Then

$$\begin{aligned} \mathbb{I}_\alpha(\mathbf{s}_n) &= \mathbf{I}(s_{\alpha_n} - s_{\alpha_{n-1}}) \mathbf{I}(s_{\alpha_{n-1}} - s_{\alpha_{n-2}}) \cdots \mathbf{I}(s_{\alpha_2} - s_{\alpha_1}) \\ &= \mathbf{I}(\hat{s}_{\hat{\alpha}_n} - \hat{s}_{\hat{\alpha}_{n-1}}) \mathbf{I}(\hat{s}_{\hat{\alpha}_{n-1}} - \hat{s}_{\hat{\alpha}_{n-2}}) \cdots \mathbf{I}(\hat{s}_{\hat{\alpha}_2} - \hat{s}_{\hat{\alpha}_1}) = \mathbb{I}_{\hat{\alpha}}(\hat{\mathbf{s}}_n). \end{aligned}$$

It follows that

$$\begin{aligned} &p(\mathbf{x}_1, \dots, \mathbf{x}_{i-1}, \mathbf{y}, \mathbf{x}_i, \dots, \mathbf{x}_{n-1}) \\ &= \sum_{\alpha \in A_n} \int_{\Omega^n} \mathbb{I}_\alpha(\mathbf{s}_n) K_\alpha(\mathbf{s}_n) \mathbf{x}_1(s_1) \cdots \mathbf{x}_{i-1}(s_{i-1}) \mathbf{y}(s_i) \mathbf{x}_i(s_{i+1}) \cdots \mathbf{x}_{n-1}(s_n) d\mathbf{s}_n \\ &= \sum_{\alpha \in A_n} \int_{\Omega^n} \mathbb{I}_{\hat{\alpha}}(\hat{\mathbf{s}}_n) K_\alpha(\mathbf{s}_n) \mathbf{x}_1(\hat{s}_1) \cdots \mathbf{x}_{n-1}(\hat{s}_{n-1}) \mathbf{y}(\hat{s}_n) d\hat{\mathbf{s}}_n \\ &= \sum_{\hat{\alpha} \in A_n} \int_{\Omega^n} \mathbb{I}_{\hat{\alpha}}(\hat{\mathbf{s}}_n) \hat{K}_{\hat{\alpha}}(\hat{\mathbf{s}}_n) \mathbf{x}_1(\hat{s}_1) \cdots \mathbf{x}_{n-1}(\hat{s}_{n-1}) \mathbf{y}(\hat{s}_n) d\hat{\mathbf{s}}_n =: \hat{p}(\mathbf{x}_1, \dots, \mathbf{x}_{n-1}, \mathbf{y}) \end{aligned}$$

□

Finally, the following lemma shows how the last factor in a homogeneous polynomial may be substituted for another homogeneous polynomial.

Lemma 12. For $\Omega = [a, b] \subseteq \mathbb{R}$ and $n \in \mathbb{N}$, let

$$p(\mathbf{x}_1, \dots, \mathbf{x}_n) = \sum_{\alpha \in A_n} \int_{\Omega^n} \mathbb{I}_\alpha(\mathbf{s}_n) K_\alpha(\mathbf{s}_n) \mathbf{x}_1(s_1) \cdots \mathbf{x}_n(s_n) d\mathbf{s}_n,$$

for $\mathbf{x}_1, \dots, \mathbf{x}_n \in L_2[\Omega]$. For $d \in \mathbb{N}$, let $\mathcal{P} := \mathcal{R}_1 \otimes \cdots \otimes \mathcal{R}_d$, where the 2-PI operators $\mathcal{R}_j$ are defined by parameters $R_{j,1}, R_{j,2} \in \mathbb{R}[s, \theta]$ for $j \in \{1, \dots, d\}$, so that

$$(\mathcal{R}_j \mathbf{x})(s) = \int_a^s R_{j,1}(s, \theta_j) \mathbf{x}(\theta) d\theta + \int_s^b R_{j,2}(s, \theta_j) \mathbf{x}(\theta) d\theta.$$

Then, for any $\mathbf{x}_1, \dots, \mathbf{x}_{n-1} \in L_2[\Omega]$ and $\mathbf{y} \in L_2[\Omega^d]$,

$$p(\mathbf{x}_1, \dots, \mathbf{x}_{n-1}, \mathcal{P}\mathbf{y}) = \sum_{\delta \in A_{n+d-1}} \int_{\Omega^{n+d-1}} \int_{\Omega^d} \mathbb{I}_\delta(\mathbf{s}_{n-1}, \boldsymbol{\theta}_d) C_\delta(\mathbf{s}_{n-1}, \boldsymbol{\theta}_d) \mathbf{x}(\mathbf{s}_{n-1}) \mathbf{y}(\boldsymbol{\theta}_d) d\boldsymbol{\theta}_d d\mathbf{s}_{n-1},$$

where $\mathbf{x}(\mathbf{s}_{n-1}) := \mathbf{x}_1(s_1) \cdots \mathbf{x}_{n-1}(s_{n-1})$, and where $C_\delta := \sum_{\alpha \in A_d} C_\delta^\alpha$ is defined as

$$C_\delta^\alpha(\zeta_{n+d-1}) = \sum_{k=\sigma_{\underline{\imath}}(\delta)}^{\sigma_{\bar{\imath}}(\delta)-1} \int_{\zeta_{\delta_k}}^{\zeta_{\delta_{k+1}}} K_\alpha(\zeta_{n-1}, s_n) \prod_{j=1}^d \left[R_{j, \gamma(j,k)}(s_n, \zeta_{n+j-1}) \right] ds_n,$$

where $\underline{\imath} := \alpha_{\sigma_n(\alpha)-1}$ and $\bar{\imath} := \alpha_{\sigma_n(\alpha)+1}$, and where

$$\gamma(j, k) := \begin{cases} 1, & \sigma_j(\delta) \leq k, \\ 2, & \sigma_j(\delta) > k, \end{cases} \quad j \in \{n, \dots, n+d-1\}, k \in \{1, \dots, n+d-1\}.$$

Proof. For ease of notation, let $\mathbf{x}(\mathbf{s}_{n-1}) = \mathbf{x}_1(s_1) \cdots \mathbf{x}_{n-1}(s_{n-1})$ and $p(\mathbf{x}, \mathbf{y}) \leftarrow p(\mathbf{x}_1, \dots, \mathbf{x}_{n-1}, \mathbf{y})$, so that

$$p(\mathbf{x}, \mathcal{P}\mathbf{y}) = \sum_{\alpha \in A_n} p_\alpha(\mathbf{x}, \mathcal{P}\mathbf{y}) := \sum_{\alpha \in A_n} \int_{\Omega^{n-1}} \int_{\Omega} \mathbb{I}_\alpha(\mathbf{s}_n) K_{k,\alpha}(\mathbf{s}_n) \mathbf{x}(\mathbf{s}_{n-1})(\mathcal{P}\mathbf{y})(s_n) ds_n d\mathbf{s}_{n-1}$$

Then, for any $\alpha \in A_n$, we have by definition of the tensor-PI operator $\mathcal{P}$ that

$$\begin{aligned} p_\alpha(\mathbf{x}, \mathcal{P}\mathbf{y}) &:= \int_{\Omega^{n-1}} \int_{\Omega} \mathbb{I}_\alpha(\mathbf{s}_n) K_{k,\alpha}(\mathbf{s}_n) \mathbf{x}(\mathbf{s}_{n-1})(\mathcal{P}\mathbf{y})(s_n) ds_n d\mathbf{s}_{n-1} \\ &= \int_{\Omega^{n-1}} \int_{\Omega} \left(\mathbb{I}_\alpha(\mathbf{s}_n) K_{k,\alpha}(\mathbf{s}_n) \sum_{\beta \in \{1,2\}^d} \int_{\Omega^d} \prod_{j=1}^d [\mathbf{I}_{\beta_j}(s_n, \theta_j) R_{j,\beta_j}(s_n, \theta_j)] \mathbf{y}(\boldsymbol{\theta}_d) d\boldsymbol{\theta}_d \right) ds_n \mathbf{x}(\mathbf{s}_{n-1}) d\mathbf{s}_{n-1} \\ &= \int_{\Omega^{n-1}} \int_{\Omega^d} \left(\sum_{\beta \in \{1,2\}^d} \int_{\Omega} \mathbb{I}_\alpha(\mathbf{s}_n) K_{k,\alpha}(\mathbf{s}_n) \prod_{j=1}^d [\mathbf{I}_{\beta_j}(s_n, \theta_j) R_{j,\beta_j}(s_n, \theta_j)] ds_n \right) \mathbf{x}(\mathbf{s}_{n-1}) \mathbf{y}(\boldsymbol{\theta}_d) d\boldsymbol{\theta}_d d\mathbf{s}_{n-1}. \end{aligned}$$

By Lemma 4, we can expand this integral into a sum over A_{n+d-1} as

$$\begin{aligned} p_\alpha(\mathbf{x}, \mathcal{P}\mathbf{y}) &= \sum_{\delta \in A_{n+d-1}} \int_{\Omega^{n-1}} \int_{\Omega^d} \mathbb{I}_\delta(\mathbf{s}_{n-1}, \boldsymbol{\theta}_d) \\ &\quad \left(\sum_{\beta \in \{1,2\}^d} \int_{\Omega} \mathbb{I}_\alpha(\mathbf{s}_n) K_{k,\alpha}(\mathbf{s}_n) \prod_{j=1}^d [\mathbf{I}_{\beta_j}(s_n, \theta_j) R_{j,\beta_j}(s_n, \theta_j)] ds_n \right) \mathbf{x}(\mathbf{s}_{n-1}) \mathbf{y}(\boldsymbol{\theta}_d) d\boldsymbol{\theta}_d d\mathbf{s}_{n-1} \\ &= \sum_{\delta \in \hat{A}_{n+d-1}^\alpha} \int_{\Omega^{n-1}} \int_{\Omega^d} \mathbb{I}_\delta(\mathbf{s}_{n-1}, \boldsymbol{\theta}_d) \\ &\quad \left(\sum_{\beta \in \{1,2\}^d} \int_{\Omega} \mathbb{I}_\alpha(\mathbf{s}_n) K_{k,\alpha}(\mathbf{s}_n) \prod_{j=1}^d [\mathbf{I}_{\beta_j}(s_n, \theta_j) R_{j,\beta_j}(s_n, \theta_j)] ds_n \right) \mathbf{x}(\mathbf{s}_{n-1}) \mathbf{y}(\boldsymbol{\theta}_d) d\boldsymbol{\theta}_d d\mathbf{s}_{n-1} \end{aligned}$$

where $\hat{A}_{n+d-1}^\alpha \subseteq A_{n+d-1}$ is the set of those permutations of $(1, \dots, n+d-1)$ that preserve the ordering from α ,

$$\hat{A}_{n+d-1}^\alpha := \left\{ \delta \in A_{n+d-1} \mid \sigma_k(\alpha) \leq \sigma_j(\alpha) \Rightarrow \sigma_k(\delta) \leq \sigma_j(\delta), \forall k, j \in \{1, \dots, n-1\} \right\}.$$

Introducing the variables $\boldsymbol{\zeta}_{n+d-1} := (\mathbf{s}_{n-1}, \boldsymbol{\theta}_d)$ and defining $\mathbf{w}(\boldsymbol{\zeta}_{n+d-1}) = \mathbf{x}(\mathbf{s}_{n-1}) \mathbf{y}(\boldsymbol{\theta}_d)$, the expression then becomes

$$\begin{aligned} p_\alpha(\mathbf{x}, \mathcal{P}\mathbf{y}) &= \sum_{\delta \in \hat{A}_{n+d-1}^\alpha} \int_{\Omega^{n+d-1}} \mathbb{I}_\delta(\boldsymbol{\zeta}_{n+d-1}) \\ &\quad \left(\sum_{\beta \in \{1,2\}^d} \int_{\Omega} \mathbf{I}(s_n - \zeta_{\underline{l}}) \mathbf{I}(\zeta_{\bar{l}} - s_n) K_{k,\alpha}(\boldsymbol{\zeta}_{n-1}, s_n) \prod_{j=1}^d [\mathbf{I}_{\beta_j}(s_n, \zeta_{n+j-1}) R_{j,\beta_j}(s_n, \zeta_{n+j-1})] ds_n \right) \\ &\quad \mathbf{w}(\boldsymbol{\zeta}_{n+d-1}) d\boldsymbol{\zeta}_{n+d-1} \end{aligned}$$

where $\underline{l} := \alpha_{\sigma_n(\alpha)-1}$ and $\bar{l} := \alpha_{\sigma_n(\alpha)+1}$. Here, for any $\delta \in \hat{A}_{n+d-1}^\alpha$, we can split the innermost integral as

$$\begin{aligned} &\sum_{\beta \in \{1,2\}^d} \int_{\Omega} \mathbf{I}(s_i - \zeta_{\underline{l}}) \mathbf{I}(\zeta_{\bar{l}} - s_i) K_{k,\alpha}(\boldsymbol{\zeta}_{n-1}, s_n) \prod_{j=1}^d [\mathbf{I}_{\beta_j}(s_n, \zeta_{n+j-1}) R_{j,\beta_j}(s_n, \zeta_{n+j-1})] ds_n \\ &= \sum_{\beta \in \{1,2\}^d} \sum_{k=0}^{n+d-1} \int_{\zeta_{\delta_k}}^{\zeta_{\delta_{k+1}}} \mathbf{I}(s_i - \zeta_{\underline{l}}) \mathbf{I}(\zeta_{\bar{l}} - s_i) K_{k,\alpha}(\boldsymbol{\zeta}_{n-1}, s_n) \prod_{j=1}^d [\mathbf{I}_{\beta_j}(s_n, \zeta_{n+j-1}) R_{j,\beta_j}(s_n, \zeta_{n+j-1})] ds_n \\ &= \sum_{\beta \in \{1,2\}^d} \sum_{k=\sigma_{\underline{l}}(\delta)}^{\sigma_{\bar{l}}(\delta)-1} \int_{\zeta_{\delta_k}}^{\zeta_{\delta_{k+1}}} K_{k,\alpha}(\boldsymbol{\zeta}_{n-1}, s_n) \prod_{j=1}^d [\mathbf{I}_{\beta_j}(s_n, \zeta_{n+j-1}) R_{j,\beta_j}(s_n, \zeta_{n+j-1})] ds_n \end{aligned}$$

where we set $\delta_0 := 0$ and $\delta_{n+d} := n+d$, and define $\zeta_0 := a$ and $\zeta_{n+d} = b$. Finally, defining

$$\gamma(j, k) := \begin{cases} 1, & \sigma_j(\delta) \leq k, \\ 2, & \sigma_j(\delta) > k, \end{cases} \quad j \in \{n, \dots, n+d-1\}, k \in \{1, \dots, n+d-1\}.$$

we get

$$\begin{aligned} & \sum_{\beta \in \{1,2\}^d} \int_{\Omega} \mathbf{I}(s_i - \zeta_{\underline{i}}) \mathbf{I}(\zeta_{\underline{i}} - s_i) K_{k,\alpha}(\zeta_{n-1}, s_n) \prod_{j=1}^d \left[\mathbf{I}_{\beta_j}(s_n, \zeta_{n+j-1}) R_{j,\beta_j}(s_n, \zeta_{n+j-1}) \right] ds_n \\ &= \sum_{k=\sigma_{\underline{i}}(\delta)}^{\sigma_{\bar{i}}(\delta)-1} \int_{\zeta_{\delta_k}}^{\zeta_{\delta_{k+1}}} K_{k,\alpha}(\zeta_{n-1}, s_n) \prod_{j=1}^d \left[R_{j,\gamma(j,k)}(s_n, \zeta_{n+j-1}) \right] ds_n =: C_{\delta}(\zeta_{n+d-1}) \end{aligned}$$

We find

$$\begin{aligned} p_{\alpha}(\mathbf{x}, \mathcal{P}\mathbf{y}) &= \sum_{\delta \in \hat{A}_{n+d-1}^{\alpha}} \int_{\Omega^{n+d-1}} \mathbb{I}_{\delta}(\zeta_{n+d-1}) C_{\delta}^{\alpha}(\zeta_{n+d-1}) \mathbf{w}(\zeta_{n+d-1}) d\zeta_{n+d-1} \\ &= \sum_{\delta \in A_{n+d-1}} \int_{\Omega^{n+d-1}} \mathbb{I}_{\delta}(\zeta_{n+d-1}) C_{\delta}(\zeta_{n+d-1}) \mathbf{w}(\zeta_{n+d-1}) d\zeta_{n+d-1}, \end{aligned}$$

and therefore

$$p(\mathbf{x}, \mathcal{P}\mathbf{y}) = \sum_{\alpha \in A_d} p_{\alpha}(\mathbf{x}, \mathcal{P}\mathbf{y}) = \sum_{\delta \in A_{n+d-1}} \int_{\Omega^{n+d-1}} \mathbb{I}_{\delta}(\zeta_{n+d-1}) C_{\delta}(\zeta_{n+d-1}) \mathbf{w}(\zeta_{n+d-1}) d\zeta_{n+d-1}$$

□

Lemma 13. Let $\Omega := [a, b] \subseteq \mathbb{R}$, $n, d \in \mathbb{N}$, and $i \in \{1, \dots, n\}$. For $\mathbf{R}_j \in \mathbb{R}[s_i, \eta_j]$ for $j \in \{1, \dots, d\}$, define the 2-PI operators

$$(\mathcal{R}_j \mathbf{x}_j)(s) = \int_a^s \mathbf{R}_{j,1}(s, \eta_j) \mathbf{x}(\eta_j) d\eta_j + \int_s^b \mathbf{R}_{j,2}(s, \eta_j) \mathbf{x}(\eta_j) d\eta_j,$$

and the tensor-PI operator

$$(\mathcal{P}[\mathbf{x}_1 \otimes \dots \otimes \mathbf{x}_d])(s) := (\mathcal{R}_1 \mathbf{x}_1)(s) \cdots (\mathcal{R}_d \mathbf{x}_d)(s),$$

for $s \in [a, b]$. Then, for any $K \in \mathbb{R}[s_1, \dots, s_n]$, $\mathbf{x} \in L_2[[a, b]^{n-1}]$ and $\mathbf{y} \in L_2[[a, b]^d]$,

$$\begin{aligned} & \int_a^b \int_{s_1}^b \cdots \int_{s_{n-1}}^b \left(C(s) \mathbf{x}(s_{n \setminus \{i\}}) (\mathcal{P}\mathbf{y})(s_i) \right) ds_n \cdots ds_2 ds_1 \\ &= \sum_{\alpha \in A_{n+d-1}} \int_{[a,b]^{n+d-1}} \mathbb{I}_{\alpha}(\boldsymbol{\theta}) \mathbf{K}_{\alpha}(\boldsymbol{\theta}) \mathbf{x}(\boldsymbol{\theta}_{n-1}) \mathbf{y}_i(\boldsymbol{\theta}_{n:n+d-1}) d\boldsymbol{\theta}_{d+i-1} \end{aligned}$$

where for $\alpha \in A_{n+d-1}$ such that $\alpha_1 \leq \dots \leq \alpha_{i-1}$ and $\alpha_{i+d} \leq \dots \leq \alpha_{n+d-1}$, we have

$$\mathbf{K}_{\alpha}(\boldsymbol{\theta}) := \sum_{k \in D_{\alpha}} \int_{\theta_{\alpha_k}}^{\theta_{\alpha_{k+1}}} \left(C(\hat{s}) \prod_{j=1}^d \left[\mathbf{R}_{j,\gamma(i+j-1,k)}(s_i, \theta_{i+j-1}) \right] \right) ds_i$$

and where $\mathbf{K}_{\alpha} = 0$ for all other $\alpha \in A_{n+d-1}$, with

$$D_{\alpha} := \{k \in \{0, \dots, n+d-1\} \mid \sigma_{\alpha}(i-1) \leq k \leq \sigma_{\alpha}(i+d-1)\},$$

and where $\sigma_{\alpha} \in \mathbb{N}^{n+d-1}$ is such that $\alpha_{\sigma_{\alpha}(k)} = k$ for $k \in \{1, \dots, n+d-1\}$.

Proof. For given $\alpha \in A_n$, let $\sigma \in \mathbb{N}^n$ be such that $\alpha_{\sigma_j} = j$ for all $j \in \{1, \dots, n\}$. Define variables $\hat{s}_j = s_{\sigma_j}$

Let $\bar{j}$ be the index of the upper bound on θ_i

Substituting the expression for the tensor-PI operator into the integral, we obtain

$$\begin{aligned} & \int_{[a,b]^{i-1}} \int_a^b \int_{[a,b]^{n-i}} \mathbb{I}_{\alpha}(\mathbf{s}_i, \boldsymbol{\eta}_{n-i}) K_{k,\alpha}(\mathbf{s}_i, \boldsymbol{\eta}_{n-i}) \mathbf{x}(\mathbf{s}_{i-1}) (\mathcal{P}\mathbf{y})(s_i) \mathbf{z}(\boldsymbol{\theta}_{n-i}) d\boldsymbol{\eta}_{n-i} ds_i d\mathbf{s}_{i-1} \\ &= \int_{\Omega^{i-1}} \int_{\Omega^{n-i}} \int_{\Omega} \left(\mathbb{I}_{\alpha}(\mathbf{s}_i, \boldsymbol{\eta}_{n-i}) K_{k,\alpha}(\mathbf{s}_i, \boldsymbol{\eta}_{n-i}) \right. \\ & \quad \left. \sum_{\beta \in \{1,2\}^d} \int_{\Omega^d} \prod_{j=1}^d \left[\mathbf{I}_{\beta_j}(s_i, \theta_j) R_{j,\beta_j}(s_i, \theta_j) \right] \mathbf{y}_j(\boldsymbol{\theta}_d) d\boldsymbol{\theta}_d \right) \mathbf{x}(\mathbf{s}_{i-1}) \mathbf{z}(\boldsymbol{\eta}_{n-i}) ds_i d\boldsymbol{\eta}_{n-i} d\mathbf{s}_{i-1} \\ &= \int_{\Omega^{i-1}} \int_{\Omega^d} \int_{\Omega^{n-i}} \left(\sum_{\beta \in \{1,2\}^d} \int_{\Omega} \mathbb{I}_{\alpha}(\mathbf{s}_{i-1}, s_i, \boldsymbol{\eta}_{n-i}) K_{k,\alpha}(\mathbf{s}_i, \boldsymbol{\eta}_{n-i}) \prod_{j=1}^d \left[\mathbf{I}_{\beta_j}(s_i, \theta_j) R_{j,\beta_j}(s_i, \theta_j) \right] ds_i \right. \\ & \quad \left. \mathbf{x}(\mathbf{s}_{i-1}) \mathbf{y}_j(\boldsymbol{\theta}_d) \mathbf{z}(\boldsymbol{\eta}_{n-i}) d\boldsymbol{\eta}_{n-i} d\boldsymbol{\theta}_d d\mathbf{s}_{i-1} \right) \end{aligned}$$

By Lemma 4, we can express this as

$$\begin{aligned}
& \int_{[a,b]^{i-1}} \int_a^b \int_{[a,b]^{n-i}} \mathbb{I}_\alpha(\mathbf{s}_i, \boldsymbol{\eta}_{n-i}) K_{k,\alpha}(\mathbf{s}_i, \boldsymbol{\eta}_{n-i}) \mathbf{x}(\mathbf{s}_{i-1}) (\mathcal{P}\mathbf{y})(s_i) \mathbf{z}(\boldsymbol{\eta}_{n-i}) d\boldsymbol{\eta}_{n-i} ds_i ds_{i-1} \\
&= \sum_{\delta \in A_{n+d-1}} \int_{\Omega^{i-1}} \int_{\Omega^d} \int_{\Omega^{n-i}} \mathbb{I}_\delta(\mathbf{s}_{i-1}, \boldsymbol{\theta}_d, \boldsymbol{\eta}_{n-i}) \\
&\quad \left(\sum_{\beta \in \{1,2\}^d} \int_{\Omega} \mathbb{I}_\alpha(\mathbf{s}_{i-1}, s_i, \boldsymbol{\eta}_{n-i}) K_{k,\alpha}(\mathbf{s}_i, \boldsymbol{\eta}_{n-i}) \prod_{j=1}^d [\mathbf{I}_{\beta_j}(s_i, \theta_j) R_{j,\beta_j}(s_i, \theta_j)] ds_i \right) \\
&\quad \mathbf{x}(\mathbf{s}_{i-1}) \mathbf{y}_j(\boldsymbol{\theta}_d) \mathbf{z}(\boldsymbol{\eta}_{n-i}) d\boldsymbol{\eta}_{n-i} d\boldsymbol{\theta}_d ds_{i-1} \\
&= \sum_{\delta \in \hat{A}_{n+d-1}^{i,d,\alpha}} \int_{\Omega^{i-1}} \int_{\Omega^d} \int_{\Omega^{n-i}} \mathbb{I}_\delta(\mathbf{s}_{i-1}, \boldsymbol{\theta}_d, \boldsymbol{\eta}_{n-i}) \\
&\quad \left(\sum_{\beta \in \{1,2\}^d} \int_{\Omega} \mathbf{I}(s_i - s_{\alpha_{\sigma_\alpha(i)-1}}) \mathbf{I}(\eta_{\alpha_{\sigma_\alpha(i)+1}-i} - s_i) K_{k,\alpha}(\mathbf{s}_i, \boldsymbol{\eta}_{n-i}) \prod_{j=1}^d [\mathbf{I}_{\beta_j}(s_i, \theta_j) R_{j,\beta_j}(s_i, \theta_j)] ds_i \right) \\
&\quad \mathbf{x}(\mathbf{s}_{i-1}) \mathbf{y}_j(\boldsymbol{\theta}_d) \mathbf{z}(\boldsymbol{\eta}_{n-i}) d\boldsymbol{\eta}_{n-i} d\boldsymbol{\theta}_d ds_{i-1}
\end{aligned}$$

where $\hat{A}_{n+d-1}^{i,d,\alpha} \subseteq A_{n+d-1}$ is the set of those permutations of $(1, \dots, n+d-1)$ that preserve the ordering from α ,

$$\hat{A}_{n+d-1}^{i,d,\alpha} := \left\{ \delta \in A_{n+d-1} \mid \begin{array}{l} \sigma_\alpha(k) \leq \sigma_\alpha(j) \Rightarrow \sigma_\delta(k) \leq \sigma_\delta(j), \forall k, j \in \{1, \dots, i-1\} \\ \sigma_\alpha(k) \leq \sigma_\alpha(j) \Rightarrow \sigma_\delta(k+d-1) \leq \sigma_\delta(j+d-1), \forall k, j \in \{i+1, \dots, n\} \end{array} \right\}.$$

Relabeling the variables as $\boldsymbol{\theta}_{n+d-1} \leftarrow (\mathbf{s}_{i-1}, \boldsymbol{\theta}_d, \boldsymbol{\eta}_{n-i})$, and defining

$$\begin{aligned}
\hat{K}(\boldsymbol{\theta}_{n+d-1}, s_i) &:= K(\theta_1, \dots, \theta_{i-1}, s_i, \theta_{i+d}, \dots, \theta_{n+d-1}) && \leftarrow K(\mathbf{s}_{i-1}, s_i, \boldsymbol{\eta}_{n-i}), \\
\mathbf{w}(\boldsymbol{\theta}_{n+d-1}) &:= \mathbf{x}(\theta_1, \dots, \theta_{i-1}) \mathbf{y}(\theta_i, \dots, \theta_{i+d-1}) \mathbf{z}(\theta_{i+d}, \dots, \theta_{n+d-1}) && \leftarrow \mathbf{x}(\mathbf{s}_{i-1}) \mathbf{y}(\boldsymbol{\theta}_d) \mathbf{z}(\boldsymbol{\eta}_{n-i}),
\end{aligned}$$

this can be expressed as

$$\begin{aligned}
& \int_{[a,b]^{i-1}} \int_a^b \int_{[a,b]^{n-i}} \mathbb{I}_\alpha(\mathbf{s}_i, \boldsymbol{\eta}_{n-i}) K_{k,\alpha}(\mathbf{s}_i, \boldsymbol{\eta}_{n-i}) \mathbf{x}(\mathbf{s}_{i-1}) (\mathcal{P}\mathbf{y})(s_i) \mathbf{z}(\boldsymbol{\eta}_{n-i}) d\boldsymbol{\eta}_{n-i} ds_i ds_{i-1} \\
&= \sum_{\delta \in \hat{A}_{n+d-1}^{i,d,\alpha}} \int_{\Omega^{n+d-1}} \mathbb{I}_\delta(\boldsymbol{\theta}_{n+d-1}) \\
&\quad \left(\sum_{\beta \in \{1,2\}^d} \int_{\Omega} \mathbf{I}(s_i - \theta_{\alpha_{\sigma_\alpha(i)-1}}) \mathbf{I}(\theta_{\alpha_{\sigma_\alpha(i)+1}} - s_i) \hat{K}_{k,\alpha}(\boldsymbol{\theta}_{n+d-1}, s_i) \prod_{j=1}^d [\mathbf{I}_{\beta_j}(s_i, \theta_{j+i-1}) R_{j,\beta_j}(s_i, \theta_{j+i-1})] ds_i \right) \mathbf{w}(\boldsymbol{\theta}_{n+d-1}) d\boldsymbol{\theta}_{n+d-1}
\end{aligned}$$

Here, for any $\delta \in \hat{A}_{n+d-1}^{i,d,\alpha}$, we can split the innermost integral as

$$\begin{aligned}
& \sum_{\beta \in \{1,2\}^d} \int_{\Omega} \mathbf{I}(s_i - \theta_{i-1}) \mathbf{I}(\theta_{i+d} - s_i) \hat{K}_{k,\alpha}(\boldsymbol{\theta}_{n+d-1}, s_i) \prod_{j=1}^d [\mathbf{I}_{\beta_j}(s_i, \theta_{j+i-1}) R_{j,\beta_j}(s_i, \theta_{j+i-1})] ds_i \\
&= \sum_{\beta \in \{1,2\}^d} \sum_{k=0}^{n+d-1} \int_{\theta_{\delta_k}}^{\theta_{\delta_{k+1}}} \mathbf{I}(s_i - \theta_{i-1}) \mathbf{I}(\theta_{i+d} - s_i) \hat{K}_{k,\alpha}(\boldsymbol{\theta}_{n+d-1}, s_i) \prod_{j=1}^d [\mathbf{I}_{\beta_j}(s_i, \theta_{j+i-1}) R_{j,\beta_j}(s_i, \theta_{j+i-1})] ds_i \\
&= \sum_{\beta \in \{1,2\}^d} \sum_{k \in D_\delta} \int_{\theta_{\delta_k}}^{\theta_{\delta_{k+1}}} \hat{K}_{k,\alpha}(\boldsymbol{\theta}_{n+d-1}, s_i) \prod_{j=1}^d [\mathbf{I}_{\beta_j}(s_i, \theta_{j+i-1}) R_{j,\beta_j}(s_i, \theta_{j+i-1})] ds_i,
\end{aligned}$$

where we define $\delta_0 = 0$, $\delta_{n+d} = n+d$, $\theta_0 = a$, and $\theta_{n+d} = b$, and where we define the set

$$D_\delta := \{k \in \{0, \dots, n+d-1\} \mid \sigma_\delta(\alpha_{\sigma_\alpha(i)-1}) \leq k \leq \sigma_\delta(\alpha_{\sigma_\alpha(i+d-1)})\},$$

and where σ_α is such that $\alpha_{\sigma_\alpha(k)} = k$ for each $k \in \{0, \dots, n+d\}$. Finally, defining

$$\gamma(j, k) := \begin{cases} 1, & \sigma(j) \leq k, \\ 2, & \sigma(j) > k, \end{cases} \quad j \in \{i, \dots, i+d-1\}, k \in \{1, \dots, n+d-1\}.$$

we get

$$\begin{aligned}\mathbb{I}_\alpha(\theta) &= \sum_{\beta \in \{1,2\}^d} \int_a^b \left(\mathbf{I}_1(s_i, \theta_{i-1}) \mathbf{I}_1(\theta_{i+d}, s_i) \mathbf{C}(\hat{s}) \prod_{j=1}^d \left[\mathbf{I}_{\beta_j}(s_i, \theta_{i+j-1}) \mathbf{R}_{j, \beta_j}(s_i, \theta_{i+j-1}) \right] \right) ds_i \\ &= \sum_{k \in D_\alpha} \mathbb{I}_\alpha(\theta) \int_{\theta_{\alpha_k}}^{\theta_{\alpha_k+1}} \left(\mathbf{C}(\hat{s}) \prod_{j=1}^d \left[\mathbf{R}_{j, \gamma(i+j-1, k)}(s_i, \theta_{i+j-1}) \right] \right) ds_i =: \mathbf{K}_\alpha(\theta)\end{aligned}$$

Substituting this relation back into (??), we find

$$\begin{aligned}& \int_a^b \int_{s_1}^b \cdots \int_{s_{n-1}}^b \left(\mathbf{C}(s) \mathbf{y}_1(s_1) \cdots \mathbf{y}_{i-1}(s_{i-1}) (\mathcal{P}\mathbf{y})(s_i) \mathbf{y}_{i+1}(s_{i+1}) \cdots \mathbf{y}_n(s_n) \right) ds_n \cdots ds_2 ds_1 \\ &= \sum_{\alpha \in \hat{A}_{n+d-1}^{i,d}} \int_{[a,b]^{n+d-1}} \mathbb{I}_\alpha(\theta) \mathbf{K}_\alpha(\theta) \mathbf{w}(\theta) d\theta = \mathcal{K} \mathbf{w}.\end{aligned}$$

□

To reduce notation, consider $\alpha = (1, \dots, n)$. For the innermost integral, we remark that if $\mathcal{R}_n = \mathcal{R}_{n,1} \otimes \cdots \otimes \mathcal{R}_{n,d_n}$, we can use the `quad2lin.term` function (or a modification thereof) to compute $\mathbf{C}_\beta^{(n)}$ such that

$$\begin{aligned}& \int_{\theta_{n-1}}^b \mathbf{K}_\alpha(\theta_1, \dots, \theta_n) (\mathcal{R}_n \mathbf{x}^{\otimes d_n})(\theta_{n-1}) d\theta_n \\ &= \sum_{\beta^{(n)} \in A_{d_n}} \int_{\theta_{n-1}}^b \int_{\theta_{n, \beta_1^{(n)}}}^b \cdots \int_{\theta_{n, \beta_{d_n}^{(n)}}}^b \mathbf{C}_{\beta^{(n)}}^{(n)}(\theta_1, \dots, \theta_{n,1}, \dots, \theta_{n,d_n}) \mathbf{x}(\theta_{n,1}) \cdots \mathbf{x}(\theta_{n,d_n}) d\theta_{n, \beta_{d_n}^{(n)}} \cdots d\theta_{n, \beta_1^{(n)}}.\end{aligned}$$

This still leaves an integral $\int_{\theta_{\alpha_{n-1}}}^b$. However, we can switch the order of integration. In particular, for any $\mathbf{y}(\theta_n)$, we have

$$\begin{aligned}& \int_{\theta_{n-2}}^b \int_{\theta_{n-1}}^b \mathbf{C}_{\beta^{(n)}}^{(n)}(\theta_1, \dots, \theta_n) (\mathcal{R}_{n-1} \mathbf{x}^{\otimes d_{n-1}})(\theta_{n-1}) \mathbf{y}(\theta_n) d\theta_{n, \beta_1^{(n)}} d\theta_{n-1} \\ &= \int_{\theta_{n-2}}^b \int_{\theta_{n-2}}^{\theta_{n, \beta_1^{(n)}}} \mathbf{C}_{\beta^{(n)}}^{(n)}(\theta_1, \dots, \theta_n) (\mathcal{R}_{n-1} \mathbf{x}^{\otimes d_{n-1}})(\theta_{n-1}) d\theta_{n-1} \mathbf{y}(\theta_n) d\theta_{n, \beta_1^{(n)}}.\end{aligned}$$

Here, the integral is again of the appropriate form for us to apply `quad2lin.term`, so that if $\mathcal{R}_{n-1} = \mathcal{R}_{n-1,1} \otimes \cdots \otimes \mathcal{R}_{n-1,d_{n-1}}$, we can compute $\mathbf{C}_{\beta^{(n-1)}, \beta}^{(n-1)}$ such that

$$\begin{aligned}& \int_{\theta_{n-2}}^{\theta_{n, \beta_1^{(n)}}} \mathbf{C}_{\beta^{(n)}}^{(n)}(\theta_1, \dots, \theta_n) (\mathcal{R}_{n-1} \mathbf{x}^{\otimes d_{n-1}})(\theta_{n-1}) d\theta_{n-1} \\ &= \sum_{\beta^{(n-1)} \in A_{d_{n-1}}} \int_{\theta_{n-2}}^{\theta_{n, \beta_1^{(n)}}} \int_{\theta_{n, \beta_1^{(n-1)}}}^b \cdots \int_{\theta_{n, \beta_{d_{n-1}-1}^{(n-1)}}}^b \\ & \quad \mathbf{C}_{\beta^{(n-1)}, \beta}^{(n-1)}(\theta_1, \dots, \theta_{n-1,1}, \dots, \theta_{n-1,d_{n-1}}, \theta_{n,1}, \dots, \theta_{n,d_n}) \mathbf{x}(\theta_{n-1,1}) \cdots \mathbf{x}(\theta_{n-1,d_{n-1}}) d\theta_{n-1, \beta_{d_{n-1}-1}^{(n-1)}} \cdots d\theta_{n-1, \beta_1^{(n-1)}}.\end{aligned}$$

Applying this approach along each of the factors $\mathcal{R}_i$, we can take the composition of the functional defined by $\mathbf{K}_\alpha$ with the tensor-PI operators $\mathcal{R}_i$. A routine performing this operation has been implemented in `mtimes_functional`. Using this approach, we can then compute the Lie derivative of the Lyapunov functional $V(x)$ along a given PIE. A routine computing this derivative has been implemented in `Liediff`.

Generalized Tensor-PI Operator

In the SOS stability test, we encounter two types of tensor products of PI operators. In particular, if $\mathcal{T} : L_2[0, 1] \rightarrow L_2[0, 1]$ represents the map from fundamental state $\mathbf{x}$ to PDE state $\mathbf{u}$, then the pointwise product of $\mathbf{u}(t, s)$ with itself is expressed by the following type of tensor product,

$$\mathbf{u}(s)^2 = (\mathcal{T}\mathbf{x})(s)^2 = ((\mathcal{T} \otimes \mathcal{T})\mathbf{x}^{\otimes 2})(s)$$

where now $\mathcal{T} \otimes \mathcal{T} : L_2[[0, 1]^2] \rightarrow L_2[0, 1]$. These products appear frequently in the PIE representation, as the PDE vector field is defined by classical, pointwise polynomials. However, to parameterize the Lyapunov functional, we may use distributed polynomials in the PDE state. These would have to be expressed in terms of a “proper” tensor product of PI operators, i.e.

$$\mathbf{u}^{\otimes 2}(s, \theta) = (\mathcal{T}\mathbf{x})^{\otimes 2}(s, \theta) = (\mathcal{T}\mathbf{x})(s)(\mathcal{T}\mathbf{x})(\theta) = ((\mathcal{T} \hat{\otimes} \mathcal{T})\mathbf{x}^{\otimes 2})(s, \theta)$$

where now $\mathcal{T} \hat{\otimes} \mathcal{T} : L_2[[0, 1]^2] \rightarrow L_2[[0, 1]^2]$. To generalize this, for $j \in \{1, \dots, d\}$, $p \in \mathbb{N}^{N \times d}$ and $q \in \mathbb{N}^{N \times d}$, define the spatial domains

$$\Omega_{1,j} := [a_1, b_1]^{p_{1,j}} \times \dots \times [a_N, b_N]^{p_{N,j}}, \quad \text{and} \quad \Omega_{2,j} := [a_1, b_1]^{q_{1,j}} \times \dots \times [a_N, b_N]^{q_{N,j}}. \quad (8)$$

Let $\mathcal{P}_{i,j} : L_2^{n_j}[\Omega_{2,j}] \rightarrow L_2^{m_j}[\Omega_{1,j}]$ for $n_j, m_j \in \mathbb{N}$ and $j \in \{1, \dots, d\}$ be (tensor-)PI operators. Using the two types of tensor products, we define two types of tensor-PI operators

$$\begin{aligned} \mathcal{R} &:= \sum_{i=1}^M \mathcal{P}_{i,1} \otimes \dots \otimes \mathcal{P}_{i,d} : L_2^{n_1 \dots n_d}[\Omega_2] \rightarrow L_2^{m_1}[\Omega], \\ \hat{\mathcal{R}} &:= \sum_{i=1}^M \mathcal{P}_{i,1} \hat{\otimes} \dots \hat{\otimes} \mathcal{P}_{i,d} : L_2^{n_1 \dots n_d}[\Omega_2] \rightarrow L_2^{m_1 \dots m_n}[\Omega_1] \end{aligned}$$

where we define

$$\begin{aligned} \Omega &= [a_1, b_1] \times \dots \times [a_N, b_N], \\ \Omega_1 &:= [a_1, b_1]^{p_{1,1} + \dots + p_{1,d}} \times \dots \times [a_N, b_N]^{p_{N,1} + \dots + p_{N,d}}, \\ \Omega_2 &:= [a_1, b_1]^{q_{1,1} + \dots + q_{1,d}} \times \dots \times [a_N, b_N]^{q_{N,1} + \dots + q_{N,d}}. \end{aligned}$$

We represent both types of operator as a **tensorvar** object **R** with the following fields

- **R.ops**: $M \times d$ cell with element $\{\mathbf{i}, \mathbf{j}\}$ a **nopvar** object representing the operator $\mathcal{P}_{i,j}$;
- **R.dim**: 1×2 array specifying the matrix dimensions of the operator
- **R.vars**: $N \times 2$ **pvar** array specifying the spatial variables $s_1, \dots, s_N$ and associated dummy variables int terms of which the operator is defined;
- **R.dom**: $N \times 2$ array specifying the intervals $\text{dom}(\mathbf{i}, :) = [a_i, b_i]$ on which each of the spatial variables is defined;
- **R.depmat1**: $N \times d$ array representing the degrees $p \in \mathbb{N}^{N \times d}$, specifying the output domain of the operators, $\Omega_{1,j} := [a_1, b_1]^{p_{1,j}} \times \dots \times [a_N, b_N]^{p_{N,j}}$
- **R.depmat2**: $N \times d$ array representing the degrees $q \in \mathbb{N}^{N \times d}$, specifying the input domain of the operators, $\Omega_{2,j} := [a_1, b_1]^{q_{1,j}} \times \dots \times [a_N, b_N]^{q_{N,j}}$
- **R.type**: Logical index specifying which type of tensor-product is used. Set to **false** for $\otimes$ and **true** for $\hat{\otimes}$

Note, if **R.type=false**, then $m_1 = \dots = m_d$ must hold for the pointwise (elementwise) product of the operators to be well-defined.

This class will also allow us to represent products of states in different spatial variables, e.g. for

$$\mathcal{P} : L_2[[a, b] \times [c, d]] \rightarrow L_2[[a, b] \times [c, d]], \quad \mathcal{Q} : L_2[[a, b] \times [c, d]] \rightarrow L_2[a, b], \quad \mathcal{R} : L_2[c, d] \rightarrow L_2[[a, b] \times [c, d]]$$

then for $\mathbf{x}, \mathbf{y} \in L_2[[a, b] \times [c, d]]$ and $\mathbf{z} \in L_2[c, d]$,

$$\begin{aligned} (\mathcal{P} \otimes \mathcal{Q} \otimes \mathcal{R})(\mathbf{x} \otimes \mathbf{y} \otimes \mathbf{z})(s_1, s_2) &= (\mathcal{P}\mathbf{x})(s_1, s_2)(\mathcal{Q}\mathbf{y})(s_1)(\mathcal{R}\mathbf{z})(s_1, s_2) \\ (\mathcal{P} \hat{\otimes} \mathcal{Q} \hat{\otimes} \mathcal{R})(\mathbf{x} \otimes \mathbf{y} \otimes \mathbf{z})(s_{11}, s_{12}, s_{13}, s_{21}, s_{22}) &= (\mathcal{P}\mathbf{x})(s_{11}, s_{21})(\mathcal{Q}\mathbf{y})(s_{12})(\mathcal{R}\mathbf{z})(s_{13}, s_{22}) \end{aligned}$$

Generalized PI Functionals

In the 1D case, we parameterize a class of PI functionals as

$$\begin{aligned}\mathcal{K}\mathbf{w} &= \sum_{\alpha \in A_d} \int_a^b \cdots \int_a^b \mathbb{I}_\alpha(\theta_1, \dots, \theta_d) \mathbf{K}_\alpha(\theta_1, \dots, \theta_d) \mathbf{w}(\theta_1, \dots, \theta_d) d\theta_d \cdots d\theta_1 \\ &= \sum_{\alpha \in A_d} \int_a^b \int_{\theta_{\alpha_1}}^b \cdots \int_{\theta_{\alpha_{d-1}}}^b \mathbf{K}_\alpha(\theta_1, \dots, \theta_d) \mathbf{w}(\theta_1, \dots, \theta_d) d\theta_{\alpha_d} \cdots d\theta_{\alpha_2} d\theta_{\alpha_1}, \quad \forall \mathbf{w} \in L_2([a, b]^d)\end{aligned}$$

where $A_d \subseteq \{1, \dots, d\}^d$ denotes the set of all possible permutations of the vector $[1 \ \cdots \ d]$. We can represent these operators numerically by storing a list of values α , and the associated kernels $\mathbf{K}_\alpha$. The disadvantage of this approach, however, is that we cannot represent integrals of the form, e.g., $\int_0^1 \mathbf{K}(s) \mathbf{x}(s)^2$, corresponding to the standard inner product. To resolve this issue, define the following characteristic function,

$$\chi_{[x,y]}(z) := \begin{cases} 1, & z \in [x, y], \\ 0, & \text{else} \end{cases} \quad x, y, z \in [a, b].$$

Using this function, we can represent the polynomial semiseparable structure used in the definition of the PI functional, integrating the variable z only over the domain $[x, y] \subseteq [a, b]$. Now, to support multiplier operators, define the Dirac delta function δ such that $\int_a^b \delta(x, z) f(z) dz = f(x)$. Then, through slight abuse of notation, we overload the definition of χ such that

$$\chi_{[x,x]}(z) := \delta(x, z), \quad x, z \in [a, b].$$

Given now a set of variables $s_1, \dots, s_d \in [a, b]$, we can distinguish the different domains of integration for the various variables by defining

$$\mathbb{I}_\xi(s_1, \dots, s_d) = \chi_{[s_{\xi_1}, s_{\xi_2}]}(s_1) \cdots \chi_{[s_{\xi_{2d-1}}, s_{\xi_{2d}}]}(s_d),$$

for $\xi \in \Xi_d$, where

$$\Xi_d \subseteq \{\xi \in \mathbb{N}^{2d} \mid \xi_{2i-1} \leq \xi_{2i}, \forall i \in \{1, \dots, d\}\}.$$

Then, for a multivariate functions, $\mathbf{v}(s_1, \dots, s_N)$, where each $s_i \in \Omega_i^{d_i} = [a_i, b_i]^{d_i}$ is also multivariate, we can parameterize a class of multivariate functionals as

$$\mathcal{K}\mathbf{v} = \sum_{\xi_1 \in \Xi_1} \cdots \sum_{\xi_N \in \Xi_N} \int_{\Omega_1^{d_1}} \cdots \int_{\Omega_N^{d_N}} \mathbb{I}_{\xi_1}(s_1) \cdots \mathbb{I}_{\xi_N}(s_N) \mathbf{v}(s_1, \dots, s_N) ds_N \cdots ds_1$$

We can represent this operator as a structure with the following fields

- **dims**: 1×2 array specifying the row and column dimension of the operator.
- **vars**: $N \times 1$ array of spatial variable names, $(s_1, \dots, s_N)$. Then, we define the kernel in terms of the default variable names $(\mathbf{s1_dum1}, \dots, \mathbf{s1_dumd1}, \mathbf{s1_dum1}, \dots, \mathbf{sN_dumdN})$.
- **dom**: $N \times 2$ array specifying for each of the spatial variables the interval on which it is defined, so that $\text{dom}(\mathbf{i}, :) = [a_i, b_i]$ implies $s_i \in [a_i, b_i]$ for each $i \in \{1, \dots, N\}$.
- **intmat**: $1 \times N$ cell specifying the matrices $\{\Xi_1, \dots, \Xi_N\}$. Then, for each $k \in \{1, \dots, N\}$, $\text{intmat}\{\mathbf{k}\} = \Xi_k$ is a $m_k \times 2d_k$ array, where m_k is the number of distinct domains of integration with regards to the k th variable, and d_k is the degree of this variable in the operator. For each $i \in \{1, \dots, m_i\}$, the row $\text{intmat}\{\mathbf{k}\}(\mathbf{i}, :) = \xi_i$ specifies the interval over which each power of the k th variable is integrated. In particular, $\text{intmat}\{\mathbf{k}\}(\mathbf{i}, 2(\mathbf{j}-1)+[1, 2]) = [p, q]$ implies that s_k^j is integrated over the domain $[s_k^p, s_k^q]$. Here, we set $s_k^0 := a_k$ and $s_k^{d_k+1} = b_k$. Furthermore, if $p = q$, the integral is replaced with a multiplier, setting $s_k^j = s_k^p$.
- **params**: $M \times 1$ cell, where $M := \prod_{i=1}^N m_i$, specifying for each of the terms in the structure the kernel $\mathbf{K}_{\xi_1, \dots, \xi_N}$ of the integral in this term.

To illustrate, consider an **intop** object with fields

$$\text{depmat2} = \left[\begin{array}{c} 2 \\ 2 \\ 1 \end{array} \right] \left\{ \begin{array}{l} x \\ y \\ z \end{array} \right\} \quad \text{intmat} = \left\{ \left[\begin{array}{cccc} 0 & 3 & 0 & 1 \\ 2 & 3 & 0 & 3 \end{array} \right], \left[\begin{array}{cccc} 0 & 3 & 1 & 1 \\ 0 & 2 & 0 & 3 \end{array} \right], [0 \quad 3] \right\}$$

This object represents a functional of the form

$$\begin{aligned}
\mathcal{K}\mathbf{v} &= \int_{[a_1, b_1]^2} \chi_{[x_0, x_3]}(x_1) \chi_{[x_0, x_1]}(x_2) \left[\int_{[a_2, b_2]^2} \chi_{[y_0, y_3]}(y_1) \chi_{[y_1, y_1]}(y_2) \left(\int_{[a_3, b_3]} \chi_{[z_0, z_3]}(z_1) \mathbf{K}_{1,1,1}(x, y, z) \mathbf{v}(x, y, z) dz \right) dy \right. \\
&\quad \left. + \int_{[a_2, b_2]^2} \chi_{[y_2, x_3]}(y_1) \chi_{[y_0, y_3]}(y_2) \left(\int_{[a_3, b_3]} \chi_{[z_0, z_3]}(z_1) \mathbf{K}_{1,2,1}(x, y, z) \mathbf{v}(x, y, z) dz \right) dy \right] dx \\
&\quad + \int_{[a_1, b_1]^2} \chi_{[x_2, x_3]}(x_1) \chi_{[x_0, x_3]}(x_2) \left[\int_{[a_2, b_2]^2} \chi_{[y_0, y_3]}(y_1) \chi_{[y_1, y_1]}(y_2) \left(\int_{[a_3, b_3]} \chi_{[z_0, z_3]}(z_1) \mathbf{K}_{2,1,1}(x, y, z) \mathbf{v}(x, y, z) dz \right) dy \right. \\
&\quad \left. + \int_{[a_2, b_2]^2} \chi_{[y_2, x_3]}(y_1) \chi_{[y_0, y_3]}(y_2) \left(\int_{[a_3, b_3]} \chi_{[z_0, z_3]}(z_1) \mathbf{K}_{2,2,1}(x, y, z) \mathbf{v}(x, y, z) dz \right) dy \right] dx \\
&= \int_{a_1}^{b_1} \int_{a_1}^{x_1} \left[\int_{a_2}^{b_2} \left(\int_{a_3}^{b_3} \mathbf{K}_{1,1,1}(x, y, z) \mathbf{v}(x, y, z) dz_1 \right) dy_1 \right. \\
&\quad \left. + \int_{a_2}^{b_2} \int_{a_2}^{y_2} \left(\int_{a_3}^{b_3} \mathbf{K}_{1,2,1}(x, y, z) \mathbf{v}(x, y, z) dz_1 \right) dy_2 dy_1 \right] dx_2 dx_1 \\
&\quad + \int_{a_1}^{b_1} \int_{x_2}^{b_1} \left[\int_{a_2}^{b_2} \left(\int_{a_3}^{b_3} \mathbf{K}_{2,1,1}(x, y, z) \mathbf{v}_1(x, y, z) dz_1 \right) dy_1 \right. \\
&\quad \left. + \int_{a_2}^{b_2} \int_{a_2}^{y_2} \left(\int_0^{z_3} \mathbf{K}_{2,2,1}(x, y, z) \mathbf{v}(x, y, z) dz_1 \right) dy_2 dy_1 \right] dx_1 dx_2
\end{aligned}$$

where we use e.g. $x_0 = a_1$ and $x_{d+1} = b_1$